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Quality-Driven Next-Best-View Planning for Target-Conditioned Egocentric 3D Reconstruction

Qualitätsgetriebene Next-Best-View-Planung für zielkonditionierte egozentrische 3D-Rekonstruktion

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Abstract

Active perception couples sensing and reconstruction: what can be reconstructed depends on what a sensing action makes visible. Under a limited acquisition budget, next-best-view planning must therefore choose which feasible observation should come next. The literature reviewed in Chapter 2 treats scene coverage, one-step reconstruction-quality ranking, target-aware information criteria, and sequential coverage separately, but leaves their conjunction for target-specific egocentric reconstruction unresolved. This thesis studies that conjunction as finite candidate selection under hard validity constraints in Aria Synthetic Environments (ASE). It adapts Relative Reconstruction Improvement (RRI) to a target-cropped point–mesh objective, separates privileged task construction and oracle evaluation from the deployable actor path, and marks GT-derived target or selected-depth inputs as non-deployable controls. The primary experiment first tests whether bounded oracle lookahead provides endpoint-quality headroom over one-step oracle greedy; a learned policy is evaluated only if that headroom passes a prespecified meaningful-effect and uncertainty rule. The current implementation supports target-specific oracle scoring and selected-action replay, with rendering memory limiting scale. Actor-visible target matching, metric repeatability, a validated held-out population, headroom, and paired policy outcomes remain unestablished. The present contribution is therefore an auditable method and experimental design, not evidence of policy superiority or deployment readiness.

Zusammenfassung

Aktive Wahrnehmung beruht auf einer geometrischen Prämisse: Was rekonstruiert werden kann, hängt davon ab, was eine Sensorhandlung sichtbar macht. Bei begrenztem Aufnahmebudget muss die Next-Best-View-Planung daher entscheiden, welche zulässige Beobachtung als Nächstes erfolgen soll. Die in Kapitel 2 ausgewertete Literatur behandelt Szenenabdeckung, einstufige Rekonstruktionsqualität, zielbezogene Informationsmaße und sequenzielle Abdeckung getrennt, lässt deren Verbindung für die zielspezifische egozentrische Rekonstruktion jedoch offen. Diese Arbeit untersucht diese Verbindung als Auswahl aus einer endlichen Kandidatenmenge unter harten Zulässigkeitsbedingungen in ASE. Sie überträgt die RRI auf ein zielbeschnittenes Punkt–Mesh-Maß, trennt privilegierte Aufgabenerzeugung und Orakelevaluation vom einsatzfähigen Akteurpfad und kennzeichnet GT-basierte Ziel- oder Tiefeneingaben als nicht einsetzbare Kontrollen. Das primäre Experiment prüft zunächst, ob eine beschränkte Orakelvorausschau gegenüber einer einstufigen gierigen Orakelstrategie einen Vorteil bei der Endpunktqualität bietet; eine gelernte Strategie wird nur bewertet, wenn dieser Vorteil ein vorab festgelegtes Relevanz- und Unsicherheitskriterium erfüllt. Die aktuelle Implementierung unterstützt zielspezifische Orakelbewertung und Replay ausgewählter Aktionen, wobei der Speicherbedarf des Renderers die Skalierung begrenzt. Nicht belegt sind bislang eine beobachtungsbasierte Zielzuordnung, die Wiederholbarkeit des Qualitätsmaßes, eine validierte zurückgehaltene Studienpopulation, ein Vorteil der Orakelvorausschau und gepaarte Strategieergebnisse. Der gegenwärtige Beitrag ist daher ein prüfbarer Methoden- und Versuchsaufbau, kein Nachweis für Strategieüberlegenheit oder Einsatzreife.

Contents

Glossary and Abbreviations

Dataset

***ADT* – Aria Digital Twin**: Project Aria dataset with real-world captures and digital-twin scene annotations. ADT is adjacent to ARIA-NBV's sim-to-real context; the current experiments focus on ASE and EFM3D/ATEK exports, while ADT remains a relevant transfer surface.

***AEO* – Aria Everyday Objects**: Small-scale real-world Project Aria object dataset used by EFM3D for egocentric 3D perception evaluation. AEO is relevant as a possible sim-to-real check for ARIA-NBV ideas that are first developed on ASE mesh-supervised snippets.

***ASE* – Aria Synthetic Environments**: Large-scale synthetic indoor dataset with simulated Project Aria sensor characteristics, egocentric trajectories, and scene annotations. ARIA-NBV uses ASE snippets and the public mesh-supervised subset to generate oracle RRI labels from ground-truth meshes and semi-dense point clouds.

***CPF* – Central Pupil Frame**: Coordinate frame placed at the midpoint between the left and right eye boxes of Project Aria glasses. CPF is used by Project Aria for gaze-related quantities and should remain distinct from rig, camera, world, and PyTorch3D frames in ARIA-NBV docs.

***MPS* – Machine Perception Services**: Project Aria processing services that derive pose, mapping, gaze, hand, and related perception outputs from sensor recordings. ARIA-NBV mainly depends on MPS-style pose and semi-dense mapping products as the current reconstruction state for RRI computation.

***MTD* – Motion Trajectory Data**: Device poses over time, usually represented as a sequence of 6-DoF transformations. MTD supplies the logged egocentric trajectory used to define snippet state, current reconstruction context, and candidate-view reference poses.

***MFCD* – Multi-Frame Camera Data**: Synchronized camera streams from multiple Project Aria cameras over a temporal window. MFCD provides the egocentric image evidence consumed by EVL and aligned with pose, calibration, and semi-dense point streams.

***MSDPD* – Multi-Semi-Dense Point Data**: Semi-dense 3D point observations generated by SLAM-style processing across a snippet or trajectory window. MSDPD is the sparse observed geometry that ARIA-NBV fuses with candidate point clouds and compares against ground-truth meshes for RRI labels.

***Project Aria* – Project Aria**: Meta's egocentric research-device and tooling ecosystem for calibrated, time-aligned multimodal sensing. ARIA-NBV treats Project Aria and its MPS-style products as the actor-visible sensing contract, while ASE meshes remain offline supervision and evaluation assets.

***SSL* – SceneScript Language**: Structured language representation for indoor scene layout using primitives such as walls, doors, windows, and objects. SceneScript is relevant to ARIA-NBV as a possible semantic/global planning layer and as one source of ASE scene-structure context.

***snippet* – Snippet**: Short synchronized temporal window of Aria sensor data used as one EVL/VIN input sample. A snippet typically contains RGB or grayscale streams, poses, calibration, semi-dense points, and scene metadata that EVL lifts into a voxel grid.

***VRS* – Virtual Reality Standard**: File format used by Project Aria tooling to store multi-modal sensor recordings efficiently. VRS is part of the upstream Project Aria ecosystem, while ARIA-NBV primarily consumes ASE/ATEK-derived tensorized snippets for current experiments.

Entity

***target* – Target of Interest**: Selected entity, object crop, point, region, or surface-deficit hypothesis whose reconstruction quality should be improved. Target-conditioned ARIA-NBV variants use this target as an explicit input to candidate scoring and planning instead of optimizing only scene-level RRI.

Geometry

***DoF* – Degrees of Freedom**: Number of independent pose parameters available to a camera, object, or action representation. ARIA-NBV distinguishes full 6-DoF pose estimation from reduced 5-DoF candidate or action spaces used for practical view planning.

***frustum* – Frustum**: Truncated pyramidal camera-visible volume bounded by near and far clipping planes plus lateral field-of-view planes. Candidate-view frusta define which scene surfaces can project into a camera image and are therefore central to visibility, rendering, and RRI diagnostics.

***LUF* – Left-Up-Forward**: Camera coordinate convention whose x axis points left, y axis points up, and z axis points forward. LUF is one of the coordinate-frame conventions that must stay explicit when moving between Project Aria cameras, PyTorch3D cameras, and ARIA-NBV candidate poses.

***OBB* – Oriented Bounding Box**: 3D bounding box with arbitrary orientation, used to represent object extent more tightly than an axis-aligned box. OBBs are a natural target encoding for entity-aware ARIA-NBV because they provide center, extent, orientation, semantic class, and confidence signals.

***6DoF* – Six Degrees of Freedom**: Pose parameterization with three translational and three rotational degrees of freedom. Project Aria poses, candidate cameras, and world-to-device transforms are usually represented as 6-DoF rigid-body transforms.

Metrics

***AUC* – Area Under Curve**: Aggregate score computed by integrating a metric curve over an acquisition, threshold, or ranking axis. AUC can summarize how quickly reconstruction quality, coverage, or ranking quality improves as views are acquired or candidate thresholds change.

***CD* – Chamfer Distance**: Historical bidirectional distance family used to compare reconstructed points against reference geometry. Thesis-facing ARIA-

NBV notation uses point-mesh error D with directional components $D_{P \to M}$ and $D_{M \to P}$; older seminar material may still call this CD.

***CR* – Coverage Ratio**: Fraction of a target surface, scene, or region treated as observed under a chosen visibility or distance threshold. Coverage ratio is a useful diagnostic baseline for NBV, but ARIA-NBV treats reconstruction-quality improvement through RRI as the preferred optimization target.

***GT* – Ground Truth**: Reference data or annotations treated as the trusted target for training, validation, or evaluation. In ARIA-NBV, ground truth usually refers to ASE meshes, object annotations, poses, or labels used to compute oracle supervision and diagnostic metrics.

***RRI* – Relative Reconstruction Improvement**: Metric quantifying the relative reconstruction-quality improvement obtained by adding a candidate observation to the current reconstruction. ARIA-NBV computes RRI by comparing reconstruction error before and after fusing candidate-view geometry, usually against an ASE ground-truth mesh for oracle supervision and evaluation.

***reward* – Target Root-Gain Reward**: Quality-only immediate reward for thesis-core target-aware rollouts. It is the target-specific point-mesh error reduction normalized by the rollout-root target error; scalar motion, rule, diversity, and validity penalties are later ablations.

***target RRI* – Target-Specific RRI**: RRI computed only on the ground-truth and reconstructed geometry associated with a selected target of interest. Target-specific RRI lets the thesis compare views by how much they improve a selected object or region, even when scene-level RRI would prefer large background surfaces.

Model

***Q-CORAL* – CORAL Q-Value Decoder**: Ordinal Q-decoder ablation that discretizes continuous fitted-Q targets into ordered classes and decodes cumulative threshold outputs to a scalar conditional value through fixed return representatives.

***EFM3D* – Egocentric Foundation Model 3D**: Egocentric 3D foundation-model stack used as the frozen spatial backbone for ARIA-NBV candidate scoring. The current project uses EFM3D and its EVL architecture to expose voxel occupancy, centerness, semantic, and OBB evidence for VIN-style RRI prediction.

***EVL* – Egocentric Voxel Lifting**: EFM3D architecture that lifts synchronized egocentric observations into a gravity-aligned 3D voxel feature volume. ARIA-NBV uses EVL head outputs, pre-head features, and OBB predictions as local actor-visible evidence and target support, not as the complete frozen scene representation.

***Q__H* – Finite-Horizon Q Function**: Target-conditioned finite-candidate value family Q__theta(s,e,a,h) that scores each valid candidate for an explicit requested residual horizon h up to a configured maximum H. The thesis-core architecture is one shared horizon-conditioned scorer; dense Q__1 and exact Q__2 targets establish the recursion, horizon-indexed fitted Q supplies longer-horizon targets, and Double Q remains an optional correction for a noisy learned successor maximum.

***target-conditioned scorer* – Target-Conditioned Scorer**: VIN-style candidate scorer that receives scene state, a candidate view, and an encoding of the target of interest. The scorer predicts target-specific utility so view ranking can prioritize a selected entity or region instead of only optimizing scene-level RRI.

***VIN* – View Introspection Network**: Learned candidate-view scorer that predicts RRI or an ordinal RRI-derived utility without capturing the candidate view. ARIA-NBV adapts VIN-NBV by placing a lightweight RRI prediction head on top of frozen EVL features and candidate-pose evidence from ASE snippets.

Planning

***cost* – Acquisition Cost**: Budget consumed to acquire observations, measured by view count, path length, elapsed time, invalid-action rate, or a weighted combination. The thesis should first report cumulative root-normalized target gain, diagnostic target RRI, and acquisition cost separately, then use scalarized objectives only when the tradeoff is explicit.

***candidate* – Candidate View**: Proposed camera pose whose expected reconstruction utility is evaluated before selecting the next observation. ARIA-NBV samples candidate views around a reference pose or target, renders candidate depths from the mesh for oracle labels, and scores candidates with RRI or learned VIN predictions.

***transition* – Counterfactual Transition**: Deterministic or replayable rollout update that renders or retrieves only the selected candidate observation, backprojects it to candidate geometry, and fuses it into the accumulated reconstruction proxy.

***action set* – Finite Candidate Action Set**: Valid discrete action set for ARIA-NBV planning, obtained by masking a sampled candidate view table. The action is the candidate-table index, not a continuous pose; the selected pose is $q_t = q_{t,a_t}$. Invalid candidates are constraints, not low-quality actions.

***return* – Finite-Horizon Return**: Symbolic H-step return used by ARIA-NBV rollout evaluation and Q_H targets. The formula keeps γ explicit while the thesis-core comparison reports cumulative root-normalized target gain under equal acquisition budget.

***5DoF* – Five Degrees of Freedom**: Reduced camera-action parameterization commonly used when roll is fixed or otherwise constrained. ARIA-NBV uses 5-DoF language for candidate and planning abstractions where position and viewing direction matter while roll is not an independently optimized control dimension.

***CF+ state* – Geometry-Rich Counterfactual State**: Ablation actor state that extends the minimal counterfactual state with selected prior synthetic observations only. The executable CF-GT carrier stores rendered depth, valid mask, calibration, selected-camera pose, source identity, and causal support; implemented S1 derives an identity-start, fixed-width current-camera surface-point residual from it.

***historic state* – Historic Snippet State**: Raw actor-visible state available along the logged Project Aria/ASE trajectory before counterfactual rollout synthesis. It includes captured camera streams, calibration, timestamps, trajectory/gravity, semidense points with support or uncertainty, EVL/EFM evidence, and detected or predicted OBBs; GT mesh and GT OBBs are excluded as actor inputs.

***CF0 state* – Minimal Counterfactual Actor State**: Main thesis-core actor-visible counterfactual rollout state. It keeps the fused point proxy $\mathcal{P}_t$ as broad scene state, optional lifted image-foundation features attached to points, local root EVL evidence for target support, selected-view history, target descriptor, budget, candidate table, validity masks and reason codes, and current candidate-query features, without exposing all-candidate GT renders.

***NBV* – Next-Best-View**: Problem of selecting the next sensor viewpoint to improve an active reconstruction or inspection objective under a limited acquisition budget. In ARIA-NBV, the preferred objective is reconstruction quality improvement rather than coverage alone, with candidate views ranked by oracle or learned RRI scores.

***oracle state* – Oracle Rollout State**: Privileged state used only for ASE mesh-supervised label generation, upper-bound planning, and evaluation. It includes GT mesh and target crops, all-candidate renders, points, normals, mesh-face visibility, RRI and point-mesh error metrics, oracle scores, and GT OBBs.

***offline state* – Persisted Offline Sample State**: Compact VIN offline-store state persisted for training and diagnostics. It contains the VinSnippetView payload, candidate table and cameras, candidate count, validity or count metadata, oracle metrics as labels, optional rendered depths, compact OBBs, trajectory metadata, and selected EVL numeric fields.

***state* – Rollout State**: Family of rollout state variants used by ARIA-NBV. The actor-visible variants contain logged or replayed observations, reconstruction proxies, candidates, validity masks, target descriptors, history, and budget; the oracle variant adds privileged GT geometry and all-candidate labels for supervision and evaluation only.

***NBV MDP* – Target-Conditioned NBV MDP**: Finite-horizon Markov decision process used to define ARIA-NBV's target-conditioned rollout and fitted Q_H training contract. The actor observes only current reconstruction state, sampled candidate views, validity masks, target descriptors, history, and budget state; oracle meshes and GT crops are supervision and evaluation signals.

***mask* – Validity Mask**: Hard candidate-level feasibility mask used during candidate ranking, rollout generation, and Q_H training. The scalar mask $m_{t,i}$ gates selectable rows, while the reason code $\rho_{t,i}$ records why a row was rejected. Invalidity is represented separately from reconstruction quality and should not be encoded as the lowest RRI class.

Protocol

***GT-EVAL* – Ground-Truth Target Evaluation**: Main thesis protocol component restricting ground-truth OBBs and target mesh crops to oracle labels, target-RRI computation, matching checks, and evaluation. GT target crops are not actor-visible inputs in the main result.

***OBS-SEL* – Observed Target Selection**: Main thesis protocol component requiring target selection to use only actor-visible evidence such as predicted or tracked OBBs, class probabilities, confidence, projected area, and semidense or EVL support. Ground-truth target annotations are not visible to the selector in this protocol.

PRED-Q **– Predicted-Target Q**: Main thesis protocol component requiring the scorer or fitted Q_H model to condition on predicted or observed target descriptors rather than ground-truth target inputs. It covers target-conditioned one-step scoring and finite-candidate Q_H selection.

Reconstruction

MVS **– Multi-view Stereo**: Reconstruction family that estimates dense or semi-dense scene geometry from multiple calibrated views. MVS is part of the broader reconstruction context for NBV, although ARIA-NBV's current oracle labels are built from ASE meshes and Aria-style semi-dense point observations.

PC **– Point Cloud**: Set of 3D points representing observed scene geometry. ARIA-NBV compares current and candidate-fused point clouds against ground-truth meshes to compute oracle RRI and surface-distance diagnostics.

SLAM **– Simultaneous Localization and Mapping**: Method for estimating sensor motion while building a map of the surrounding scene. In ARIA-NBV, logged SLAM poses and semi-dense points form the current reconstruction state against which candidate-view RRI is evaluated.

track **– Track**: Temporal sequence of corresponding image-feature detections across frames, usually carrying per-frame image coordinates, timestamps, and camera IDs. Tracks can be triangulated or optimized into 3D points, making them a bridge between image evidence and the semi-dense point clouds used by ARIA-NBV.

VIO **– Visual-Inertial Odometry**: Pose-estimation method that combines visual measurements from cameras with inertial measurements from IMUs. Project Aria pose and mapping products build on visual-inertial estimation, and ARIA-NBV depends on those pose streams for snippet state and candidate frame contracts.

Representation

***Occupancy Grid* – Occupancy Grid**: Spatial grid whose cells encode whether space is occupied, free, unknown, or represented by a related occupancy probability. Occupancy-style voxel evidence appears in EVL outputs and VIN feature construction, where it helps summarize local 3D scene state for candidate scoring.

***3DGS* – 3D Gaussian Splatting**: Explicit radiance-field representation using optimized 3D Gaussian primitives for real-time novel-view synthesis. ARIA-NBV treats active 3DGS work as proposal, uncertainty, and simulator-bridge literature, not as a replacement for ASE mesh-supervised RRI.

Supervision

***oracle RRI* – Oracle RRI**: RRI label computed with privileged ground-truth geometry, used for supervised training and evaluation. The current oracle renders candidate depth maps from ASE ground-truth meshes, backprojects candidate point clouds, fuses them with the current semi-dense reconstruction, and scores the resulting surface-distance improvement.

List of Symbols

Symbol	Meaning	Key
	Oracle and Reconstruction	
RRI	Relative Reconstruction Improvement score for a candidate view.	`oracle.rri`
$\mathcal{P}$	Actor-visible point set accumulated from observed or replayed views.	`oracle.points`
$\mathcal{P}_q$	Candidate-view point contribution used in oracle or counterfactual updates.	`oracle.points_q`
$\mathcal{Q}$	Finite candidate-view set considered by the planner.	`oracle.candidates`
$\mathcal{Q}_t$	Candidate-view set available at rollout step t.	`oracle.candidates_t`
$q_{t,i}$	Candidate pose i at rollout step t.	`oracle.candidate_qti`
$\boldsymbol{c}$	World-space camera-center vector; subscripts identify a rollout root or candidate pose.	`oracle.center`
$\boldsymbol{D}_q$	Oracle-rendered or candidate-specific depth map for a proposed view.	`oracle.depth_q`
$D_{P \to M}$	Point-to-mesh directional error component.	`oracle.acc`
$D_{M \to P}$	Mesh-to-point directional error component.	`oracle.comp`
D	Aggregate point-mesh reconstruction error used by RRI definitions.	`oracle.err`
	ASE Assets	
$\mathcal{M}^{\mathrm{GT}}$	Ground-truth scene mesh used for oracle labels and evaluation.	`ase.mesh`
$\mathcal{M}_e^{\mathrm{GT}}$	Ground-truth mesh crop for target e.	`ase.mesh_target`
$\mathcal{F}^{\mathrm{GT}}$	Ground-truth mesh faces used in point-mesh distance calculations.	`ase.faces`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(t)$	Project Aria rig pose in the world frame at time t.	`ase.traj`
$\boldsymbol{T}_{\mathrm{rig}}^{\mathrm{w}}(T)$	Final rig pose used as an anchor for gravity-aligned local fields.	`ase.traj_final`
	VIN Scorer	
r	Target or scene RRI value used as a model target.	`vin.rri`
$\hat{r}$	Predicted RRI value emitted by the learned scorer.	`vin.rri_hat`
$\mathcal{L}$	Training loss for scorer or value-model objectives.	`vin.loss`
m	Candidate validity indicator used by scorer diagnostics.	`vin.cand_valid`

Symbol	Meaning	Key
	Entity and Target	
RRI_e	Target-specific Relative Reconstruction Improvement for entity e.	`entity.rri_e`
$\mathrm{RRI}^e_{t,i}$	State-relative marginal target RRI for candidate i at rollout step t.	`entity.target_rri_marginal`
$C_t^{\mathrm{RRI},e}$	Running sum of selected state-relative target RRIs.	`entity.target_rri_cumulative`
J_t^e	Running cumulative target gain normalized by rollout-root error.	`entity.target_root_gain_cumulative`
$\boldsymbol{\varphi}_e$	Actor-visible target/entity descriptor used to condition target-specific value prediction.	`entity.target_desc`
$\boldsymbol{p}_e^w$	World-space center of the selected target entity e.	`entity.center`
	Planning and Rollout	
s	Abstract planning state.	`rl.s`
a	Action index selecting a candidate row.	`rl.a`
r	Scalar reward or immediate gain.	`rl.r`
G	Finite-horizon return accumulated across rollout steps.	`rl.G`
j	Factual rollout-chain index used to preserve common trajectory heritage across selected steps.	`rl.rollout_index`
$\mathcal{M}_{\mathrm{NBV}}$	Target-conditioned finite-candidate NBV decision process.	`rl.mdp_nbv`
$\mathcal{A}(s_t)$	Valid action set available in state s_t.	`rl.action_set`
T	State-transition operator for selected candidate actions.	`rl.transition`
s_t^{hist}	Logged historic snippet state at rollout step t.	`rl.s_hist`
s_t^{off}	Persisted offline sample state used by training readers.	`rl.s_off`
s_t^{cf0}	Minimal actor-visible counterfactual state.	`rl.s_cf0`
$s_t^{\mathrm{S0\text{-}pose}}$	Implemented fixed-root pose-history state used by qh_cf0_v1.	`rl.s_pose`
$s_t^{\mathrm{cf+}}$	Geometry-enriched counterfactual successor state.	`rl.s_cf_geom`
$s_t^{\mathrm{CF\text{-}GT\text{-}carrier}}$	Implemented privileged selected-depth carrier used by qh_cfplus_gt_depth_v1.	`rl.s_cf_gt_carrier`
s_t^{oracle}	Oracle-only state used for labels, upper bounds, and evaluation.	`rl.s_oracle`
r_t^e	Target-specific reward at rollout step t.	`rl.reward_target`
	Entity and Target	

Symbol	Meaning	Key
r_t^e	Canonical target-specific reward at rollout step t.	`entity.target_reward`
	Planning and Rollout	
$G_t^{(H)}$	H-step finite-horizon return from rollout step t.	`rl.return_h`
Q_H	Finite-horizon candidate-value function.	`rl.qh`
$Q_{h,\theta,e,i}^{\text{cond}}$	Action-mask-independent conditional candidate value emitted by the scorer.	`rl.conditional_q`
$\ell_{t,i}^{\text{feas}}$	Physical or observed feasibility logit emitted by the scorer's feasibility head.	`rl.feasibility_logits`
e_k^Q	Fixed continuous fitted-Q boundary between adjacent CORAL classes.	`rl.coral_q_edge`
u_k^Q	Fixed continuous-Q representative used to decode one CORAL class.	`rl.coral_q_value`
c_n^Q	Ordinal class assigned to one continuous fitted-Q target.	`rl.coral_q_label`
γ	Return discount factor.	`rl.gamma`
H	Planning or rollout horizon length.	`rl.H`
$H_{\max}$	Maximum residual horizon admitted by a scorer/data contract.	`rl.H_max`
h	Scalar residual horizon requested for one scorer query; it must satisfy 1 <= h <= b_t <= H_max.	`rl.requested_horizon`
$m_{t,i}$	Hard validity mask for candidate i at rollout step t.	`rl.validity_mask`
$m_{t,i}^{\text{cand}}$	Materialized candidate row versus structural padding.	`rl.candidate_row_mask`
$m_{t,i}^{\text{act}}$	Authoritative physically valid action support.	`rl.action_mask`
$m_{t,i}^{Q,h}$	Availability of a finite factual value target for candidate i at requested horizon h.	`rl.q_label_mask`
$m_{t,i}^F$	Availability of a trustworthy feasibility label for candidate i.	`rl.feasibility_label_mask`
m_t^{succ}	Availability of a factual successor backup for transition t.	`rl.successor_mask`
$\ell_{t,i}^{\text{src}}$	Actor-oracle source-provenance role for candidate evidence.	`rl.source_role`
$\rho_{t,i}$	Invalid-action reason code for candidate i at rollout step t.	`rl.invalid_reason`
e_t	Selected target entity at rollout step t.	`rl.target`
b_t	Remaining acquisition budget at rollout step t.	`rl.budget`
$C(\tau)$	Acquisition cost of a selected trajectory.	`rl.acquisition_cost`

Symbol	Meaning	Key
	Shape and Size	
N_q	Number of candidate views in a candidate table.	`shape.Nq`
K	Number of ordinal bins or discrete levels when used in scorer losses.	`shape.K`
	scene	
$\boldsymbol{M}_t^{\mathrm{ray}}$	Sparse actor-visible ray-aware memory separating occupied, free, and unknown support at step t.	`scene.ray_memory_t`
	Observation and Actor State	
$\boldsymbol{F}_t^{\mathrm{DINO@pt}}$	Visibility-gated logged DINO feature bank attached to semidense or fused world points.	`obs.dino_point_bank_t`
$\boldsymbol{X}_t^{\mathrm{pt}}$	Point-token set combining geometry, support, uncertainty, history, and optional compressed logged descriptors.	`obs.point_tokens_t`
	scene	
$\boldsymbol{\Phi}_t^{\mathrm{scene}}$	Composite actor-visible scene-memory descriptor queried by the finite-candidate value model.	`scene.scene_memory_t`
$\boldsymbol{E}_0^{\mathrm{EVL\text{-}local}}$	Root local EVL evidence field used for target/candidate support reads.	`scene.evl_local`
$\omega_{t,i}^{\mathrm{EVL}}$	Candidate or target-query fraction supported by the root EVL voxel extent.	`scene.evl_support_frac`
$\boldsymbol{g}_{t,i}^{\mathrm{EVL}}$	Pooled local EVL evidence token for a target, candidate frustum, or target-candidate intersection query.	`scene.evl_support_token`
$\boldsymbol{g}_e^{\mathrm{tgt}}$	Pooled target-support descriptor for the selected target hypothesis.	`scene.target_support_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{fr}}$	Candidate-frustum support descriptor pooled from actor-visible scene memory.	`scene.frustum_support_pool`
$\boldsymbol{g}_{t,e,i}^{\mathrm{cap}}$	Pooled descriptor over the intersection between target support and candidate frustum support.	`scene.target_frustum_pool`
$\boldsymbol{g}_{t,i}^{\mathrm{ray}}$	Candidate-conditioned ray query over occupied, free, unknown, hit, target-support, and uncertainty summaries.	`scene.ray_query_ti`
RenderQuery	Abstract query operator that summarizes scene memory in a candidate camera/frustum without introducing fresh counterfactual RGB features.	`scene.render_query`
	spatial	
r_t	Reference pose for candidate-relative descriptor construction at decision step t.	`spatial.ref_pose`

Symbol	Meaning	Key
$\boldsymbol{T}^{\mathrm{rel}}_{r_t,i}$	Relative transform from the reference pose r_t to candidate camera i.	`spatial.ref_candidate_transform`
$\boldsymbol{h}^{\mathrm{pose}}_{t,i}$	Relative/local candidate pose descriptor derived from a reference pose rather than raw world coordinates.	`spatial.candidate_pose_feat`
$\boldsymbol{h}^{\mathrm{rel}}_{t,e\|i}$	Candidate-target relation descriptor in the candidate/ query local frame.	`spatial.candidate_target_rel_feat`
$\boldsymbol{e}^{\mathrm{rel}}_{a\|i}$	Query-local relative positional embedding for target, history, support, or candidate relations.	`spatial.relation_rpe`
r_e	Geometric-mean target-proxy scale derived from the target OBB semi-axis lengths.	`spatial.target_obb_scale`
$\hat{\boldsymbol{\delta}}^{e}_{j,t}$	Unit factual selected-camera displacement expressed in target-object coordinates for rollout j and step t.	`spatial.target_frame_motion_direction`
$\hat{\boldsymbol{v}}^{e}_{j,t}$	Unit selected-camera forward optical axis expressed in target-object coordinates.	`spatial.target_frame_view_direction`
$\mathscr{F}^{e}_{j,t}$	Front-facing target-proxy surface directions that project inside the calibrated selected-camera image.	`spatial.target_frame_frustum`
$\kappa^{e}_{j,t}$	Fraction of the target-centred proxy sphere geometrically supported by one selected calibrated frustum.	`spatial.target_frame_frustum_fraction`
	model	
$\boldsymbol{h}^{\mathrm{tgt}}_{e}$	Learned selected-target token built from the actor-visible target descriptor and scene support.	`model.target_token`
	spatial	
$\Omega^{\mathrm{FOV}}_{j,t}$	Intrinsic solid angle of one calibrated selected-camera pinhole field of view.	`spatial.frustum_solid_angle`
	model	
$\boldsymbol{x}_{t,i}$	Per-candidate row feature assembled from pose, relation, support, validity, provenance, and history descriptors.	`model.candidate_row`
$\boldsymbol{p}^{\mathrm{hist}}_{t,j}$	Previously selected pose j encoded from the current camera at decision state t.	`model.history_pose_feature`
$a^{\mathrm{hist}}_{t,j}$	Normalized relative age of selected pose j at decision state t; the immediate predecessor has age zero.	`model.history_relative_age`
$\boldsymbol{h}^{\mathrm{hist}}_{t}$	Fixed-width causal selected-pose history token supplied to scorer state fusion.	`model.history_token`
	Entity and Target	

Symbol	Meaning	Key
$J_e^{(H)}$	Endpoint reconstruction gain for target e over horizon H.	`entity.endpoint_gain`
$J_{e,\log}^{(H)}$	Log-scale endpoint reconstruction gain for target e.	`entity.log_gain`
Δ_{look}	Gain available to an oracle lookahead policy beyond one-step selection.	`entity.lookahead_headroom`
η_Q	Fraction of oracle lookahead gain recovered by the Q policy.	`entity.q_recovery`
$G_t^{(H)}$	Target-conditioned finite-horizon return.	`entity.return_h`
	Observation and Actor State	
$\boldsymbol{D}$	Depth observation sequence.	`obs.depth`
$\boldsymbol{n}$	Per-point or per-face normal observation.	`obs.face_normal`
$\boldsymbol{I}^{\text{rgb}}$	RGB image observation.	`obs.img_rgb`
$\mathcal{P}^{\text{cf}}$	Counterfactual point observation.	`obs.points_cf`
$\mathcal{P}^{\text{semi}}$	Semi-dense observed point set.	`obs.points_semi`
$\mathcal{P}_t$	Point set available at rollout step t.	`obs.points_t`
	Planning and Rollout	
$\mathcal{Q}_t$	Finite candidate-view table at rollout step t.	`rl.candidate_table`
$y_t^{(2,\text{exact})}$	Exact two-step fitted-Q control using factual dense successor one-step rewards.	`rl.exact_q2_target`
$\varepsilon_t^{(2)}$	Absolute learned-recursion target error against the exact two-step control.	`rl.q2_recursion_error`
	VIN Scorer	
$\boldsymbol{F}_v$	Learned value field evaluated at v.	`vin.field_v`

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1 Introduction

Active perception begins from a geometric premise: what can be reconstructed depends on what a sensing action makes visible [AWB88, Baj88]. A new view changes both the available surface evidence and the state from which later views are chosen. Next-best-view (NBV) planning turns this coupling between action, observation, and reconstruction into the repeated choice of where to look next [Jia+25, SRR03].

A spatial representation is useful here because of the decisions it supports, not because map fidelity is an end in itself. Dense-map representations preserve different queries and incur different sensing, computational, and storage trade-offs, so their adequacy depends on the downstream task [Rei+26]. For target-specific NBV, the relevant test is therefore whether the actor state preserves the distinctions needed to predict the reconstruction consequences of candidate views—not whether it reproduces the most complete possible world model.

With a limited acquisition budget, a view is "best" only relative to an objective. Coverage rewards newly observed surface, information criteria reward reduced uncertainty, and quality-driven methods evaluate the reconstruction itself [Fra+25, GML22, JLD24]. These objectives can disagree. Even "information gain" is not unique: D-, A-, and E-optimal design summarize uncertainty volume, average variance, and worst-case variance, respectively [Ros+26]. A view may reveal more of a room or reduce one uncertainty summary without exposing the occluded surface of a requested object, while a small side-step may improve that object but add little scene-wide coverage. Target conditioning therefore specifies **whose geometry** should improve; it is not generic semantic awareness.

Existing methods supply the necessary ingredients under different assumptions. VIN-NBV ranks sampled candidates by direct reconstruction improvement, object-aware 3D Gaussian Splatting conditions an information criterion on a requested object, and GenNBV and Hestia learn sequential coverage policies [Che+24, Fra+25, Jeo+26, Lu+26]. What remains unresolved is their conjunction: whether target-specific reconstruction quality contains useful multi-step structure and whether that structure can be recovered from a bounded egocentric information state.

Project Aria supplies calibrated egocentric observations, EFM3D lifts them into local 3D evidence, and ARIA Synthetic Environments (ASE) provides the privileged geometry needed for controlled target tasks and counterfactual evaluation [Eng+23, Met25, Str+24a]. This combination makes the information boundary testable. Task construction, action-set construction, hard admission, supervision, and evaluation may use privileged geometry, while the scorer consumes only the fields admitted by its declared actor-facing protocol. Scientific interpretation must account for both paths. Chapter 3 therefore treats actor visibility as a property of the complete decision process, not merely of the scorer tensor.

1.1 Aim and bounded problem

The evaluation proceeds in two stages. First, it measures whether bounded oracle lookahead yields better fixed-budget endpoint reconstruction of a requested object than one-step oracle-greedy selection. Conditional on such headroom, it then measures how much of the advantage an offline finite-horizon model recovers from actor-visible inputs. This order separates the existence of a planning opportunity from the ability of a learned model to exploit it.

At every step, each policy selects from the same finite generated candidates; hard-invalid actions are excluded, and the target, horizon, candidate support, and endpoint evaluation are matched. The current oracle tasks use geometry-valid ground-truth boxes, while actor-visible target discovery and matching remain separate research requirements. The objective adapts VIN-NBV rather than copying it: VIN-NBV defines RRI from point-cloud Chamfer distance [Fra+25], whereas ARIA-NBV evaluates a target-cropped point–mesh error after an equal acquisition budget. The resulting claims concern this bounded finite-candidate setting, not a deployable target-selection system or an objective invariant to other reconstruction protocols.

1.2 Contributions and current evidence

The thesis operationalizes this evaluation through four scientific functions:

1. It separates one-step RRI, finite-horizon return, and fixed-budget endpoint gain within target-specific reconstruction quality.
2. It defines an end-to-end information boundary across task and action construction, state updates, model inputs, supervision, and evaluation.
3. It defines candidate value relationally over target, causal state, admissible action, budget, horizon, and continuation rule.
4. It evaluates measurement, support, oracle headroom, immediate-value learning, recursion, and endpoint recovery in that order.

Before model capacity becomes the main question, the evidence needed to define the learning problem must itself be identifiable. The outcome must be repeatable, the study population and candidate support must contain the relevant opportunity, bounded lookahead must exhibit headroom, and factual replay must identify the required targets. Only after those conditions hold can a failure to recover non-myopic value be attributed primarily to representation or learning. The order makes a negative result informative: it distinguishes a measurement or support failure from a failure of immediate-value learning or finite-horizon recursion.

The present evidence supports a methodological contribution rather than a policy result. The implementation executes target-specific oracle scoring and selected-action replay, while rendering memory limits scale. Actor-visible target matching, metric repeatability, a validated held-out population, oracle-lookahead headroom, and paired policy outcomes remain unestablished. Accordingly, the thesis does not yet claim policy superiority or deployment readiness.

1.3 Research Questions

1.3.1 Research aim and evaluation logic

RQ2 joins two conditional comparisons: whether bounded oracle lookahead improves fixed-budget endpoint reconstruction relative to one-step oracle-greedy selection and, if so, how much of that advantage an actor-visible offline model recovers. RQ1 defines the outcome, RQ3 defines the admissible information, and RQ4 establishes the target, action, and replay support over which the comparison is interpretable. RQ5 and RQ6 extend the study only if the offline finite-candidate evidence warrants online or continuous-control claims.

1.3.2 RQ2 — Bounded lookahead and learned recovery

How much of the fixed-budget endpoint advantage of bounded oracle lookahead over one-step oracle-greedy selection can an offline finite-horizon value model recover from actor-visible evidence?

The paired endpoint difference under the same scene, target, candidates, validity rules, horizon, and budget defines oracle-lookahead headroom:

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

The learned comparison proceeds only if headroom passes a predeclared meaningful-effect and uncertainty rule. Because that rule and the required recovery fraction are not yet frozen, RQ2 remains prospective. If headroom is absent, the evaluated support does not expose a non-myopic advantage; this does not establish its universal absence.

1.3.3 Conditions for interpretation

1.3.3.1 RQ1 — Target-specific objective and endpoint outcome

Can target-conditioned finite-candidate NBV be evaluated through a stable, target-specific reconstruction objective under a fixed acquisition budget?

For a target e, the oracle measures reconstruction error on a frozen target crop. The primary estimand is the quality change from the root state to the endpoint after H acquisitions:

$$J_e^{(H)} = \frac{\Delta_0^e - \Delta_H^e}{\Delta_0^e + \varepsilon}$$

State-relative RRI is a one-step diagnostic, root-normalized gain supplies the training reward, and endpoint gain remains the policy outcome. A positive answer requires a frozen, repeatable metric and equal acquisition horizons; runtime, invalid actions, and oracle calls are reported separately.

1.3.3.2 RQ3 — Actor-visible target and information state

Which end-to-end target, action-support, and history protocol supports one-step and finite-horizon candidate scoring without privileged target geometry, oracle labels, or unselected candidate renders at decision time?

Ground-truth target tasks remain restricted to task construction, supervision, and evaluation. The actor-facing protocol must account for how the target instruction, candidate support, hard validity mask, selected state update, and scorer inputs are produced. The scorer receives only the declared target descriptor, causal egocentric evidence, remaining budget, and requested horizon; evaluation separately audits matching failures, ranking and calibration, and actor/oracle leakage.

1.3.3.3 RQ4 — Candidate, replay, and population support

Do the candidate generator, validity rules, rollout recipes, and replay population provide adequate and diverse support for the RQ1–RQ3 estimands?

The candidate table combines exploration and target-bearing views, records generator provenance, and excludes geometric infeasibility through a hard mask. RQ4 measures candidate-family survival, invalidity, scene and target coverage, replay and horizon support, and resource cost. Scene-disjoint splits and scene-level aggregation prevent dense sampling from masquerading as population coverage.

1.3.4 Scope extensions

1.3.4.1 RQ5 — Online discrete bridge

If offline headroom, replay support, and actor-visible scoring are established, does online interaction over the unchanged discrete candidate set and validity rules improve endpoint target gain or calibration over the offline policy?

RQ5 keeps the target, information, candidate, validity, and endpoint assumptions of RQ1–RQ4 unchanged. It is not required for the offline finite-candidate evaluation.

1.3.4.2 RQ6 — Continuous or simulator-backed control

If the finite-candidate evidence is stable, does a continuous or hierarchical target-then-pose policy provide measurable headroom over the best discrete policy under the same target-specific objective and a comparable acquisition or motion budget?

RQ6 changes the action space and feasibility assumptions and therefore requires separate evidence for support, safety, simulator realism, and comparable cost. It remains deferred; the current implementation implies no continuous, simulator, or real-device result.

1.3.5 Shared evidence constraints

Action validity, actor support, target validity, oracle-label validity, and training eligibility remain distinct; missing evidence is not encoded as low utility. Confirmatory comparisons are paired under the same scene, target, candidates, validity regime, and budget. The scene is the experimental unit, and the analysis plan freezes exclusions, aggregation, uncertainty, and decision rules before policy outcomes are inspected.

1.3.6 Research-to-evidence map

RQ	Question role	Primary evidence	Interpretation gate
RQ2	lookahead and recovery	paired greedy, lookahead, and learned-policy outcomes	meaningful headroom before recovery
RQ1	outcome measurement	target reconstruction endpoint gain	frozen repeatable metric; fixed horizon and budget
RQ3	information boundary	matching, ranking, calibration, and leakage audits	end-to-end actor-visible protocol
RQ4	population and support	candidate, replay, validity, and coverage diagnostics	scene-disjoint aggregation
RQ5	conditional extension	matched online discrete-policy evaluation	offline gates satisfied first
RQ6	deferred extension	continuous or simulator-backed evaluation	separate action space and cost comparison

Table 1: Research-question-to-evidence map.

1.4 Thesis structure

The remaining chapters follow that dependency chain. Chapter 2 derives the objective, support, temporal, and representation concepts needed to state the gap precisely. Chapter 3 constructs the experimental world by defining information access, admissible actions, state transitions, and measured outcomes. Chapter 4 then specifies the one finite-horizon value interface evaluated within that world. Chapter 5 states which evidence admits or blocks each claim. Chapter 6 reports the admitted answers in gate order; Chapter 7 attributes their implications and alternatives; and Chapter 8 synthesizes what those bounded answers establish.

2 Foundations and Related Work

The Introduction located this thesis within active perception: sensing actions change the evidence on which later inference depends [AWB88, Baj88]. Three-dimensional view planning turns that principle into a repeated choice of where to observe next [SRR03]. The unresolved dependency is not another planner implementation but a precise account of what a view is valuable **for**, which actions are available, and what information a sequential score may condition on. By the end of the chapter, the thesis question is reduced to three coupled tensions—objective, temporal, and support/state—before the experimental world is constructed.

The argument follows the dependencies of the decision problem. It first separates the next-best-view mechanism from the objective used to rank views, then distinguishes target-conditioned reconstruction quality from coverage and uncertainty proxies. It next separates candidate-view support from endpoint, transition, and human-motion feasibility before explaining why partial observability and delayed consequences require a finite-horizon information state. The chapter then derives the geometric properties a candidate scorer should preserve and positions the thesis question against the resulting literature dimensions.

2.1 Active Perception and the Next-Best-View Problem

In active perception, an observation is also an intervention on the future information state. A next-best-view (NBV) method operationalizes this idea by generating possible viewpoints, estimating the benefit of observing from each one, and selecting an action. PB-NBV makes this generate–score–select decomposition explicit for a finite candidate set [Jia+25]. The decomposition identifies three different scientific objects: the proposal mechanism determines which views can be considered, the utility determines how they are ordered, and the selection rule turns that ordering into an action.

How those objects are represented varies across NBV systems. Finite-candidate methods evaluate a bounded set of poses, whereas learned continuous policies predict camera position and orientation directly. GenNBV formulates the latter as a five-degree-of-freedom reinforcement-learning problem, while Hestia factorizes the continuous action into a look-at point followed by a camera position [Che+24, Lu+26]. A finite set makes the evaluated choices explicit and comparable; a continuous or hierarchical policy can search a broader space but couples view proposal and control more tightly.

These action formulations do not determine what makes a view useful. PB-NBV, for example, accelerates candidate evaluation through projected frontier and occupied evidence, so its proposal and scoring mechanism is tied to a coverage objective [Jia+25]. The same finite candidate table could instead be ranked by uncertainty reduction or by reconstruction error. Consequently, changing the utility changes the scientific task even when candidate generation and selection remain unchanged.

The NBV mechanism therefore answers only **how** a sensing choice is made. To interpret the choice, the utility must state **which reconstruction consequence** is valued and **for which spatial support**. This distinction leads from the general active-perception loop to the objective families compared next; only after fixing that objective can candidate support and feasibility be separated from view value [Che+24, Jia+25].

2.2 View Utility and Target-Conditioned Reconstruction

Coverage-based utilities value evidence about previously unobserved surface. SCONE estimates surface-coverage gain from occupancy and visibility fields, MACARONS anticipates coverage gain while reconstructing online, and Hestia rewards newly visible voxel faces [GML22, Gué+23, Lu+26]. These objectives connect a view to geometric completeness, but they weight every counted surface element through the coverage definition; they do not directly ask whether the reconstructed geometry became more accurate.

Information-based utilities instead value a view through its expected reduction of model uncertainty. FisherRF derives an information-gain criterion from the Fisher information of a radiance field [JLD24]. Such a criterion can prefer a view that constrains uncertain model parameters even when its immediate surface coverage is small. The benefit is model-aware exploration; the limitation is that lower parameter uncertainty and lower geometric reconstruction error are related only through assumptions about the model and the evaluated scene.

Direct quality utilities close this proxy gap by evaluating the reconstruction itself. VIN-NBV defines Relative Reconstruction Improvement as the reduction in point–mesh reconstruction error after adding a candidate observation, then learns to rank sampled query cameras by that quantity [Fra+25]. The objective measures the desired geometric consequence directly, but the learned scorer remains myopic: each candidate is valued by the improvement attributed to that next observation rather than by the sequence it may enable.

Target conditioning changes which reconstruction errors contribute to that quality. Object-centric 3DGS planning associates features and confidence with individual Gaussian primitives and gates its information score by the requested object [Jeo+26]. This mechanism demonstrates that a view can be informative for one object without being equally useful for the scene as a whole. It also exposes a separate assumption: supplying an object identifier or region for conditioning is not the same problem as discovering that object and maintaining its identity from observations.

The thesis therefore treats coverage and uncertainty as non-equivalent diagnostics and takes target-specific reconstruction quality as the endpoint of interest [Fra+25, Jeo+26]. This choice fixes what the decision should improve, but it does not determine which viewpoints can be proposed or reached. Candidate support and motion feasibility must be separated from utility before delayed consequences are embedded in a sequential decision problem [Jia+25].

2.3 Candidate-View Support and Motion Feasibility

A finite-candidate policy selects only from the pose rows in its proposed candidate table $\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_t}$. PB-NBV uses a reachability- and camera-conditioned partial hemisphere of target-facing candidates [Jia+25]. Proposal geometry therefore bounds what even a perfect scorer can select.

Target-relative coordinates provide a useful proposal prior without defining the objective. PB-NBV points candidates toward the object centre, while Hestia factorizes a viewpoint into a predicted look-at point and a camera position [Jia+25, Lu+26]. Such parameterizations concentrate support on a requested target, but an orbit or target-facing arc remains a coverage geometry, not evidence that a wearer normally moves that way. Project Aria explicitly contrasts carefully curated scanning motion with natural, unconstrained egocentric activity, so wearable plausibility must be checked against trajectory evidence rather than inferred from a target-relative construction [Eng+23].

Geometric admissibility has at least two levels. Hestia shifts a proposed camera position to its nearest collision-free endpoint, whereas Next Best Sense obtains a list of feasible candidate views but still falls back to the next-ranked view when inverse kinematics or trajectory planning fails [Lu+26, Str+24b]. A collision-free endpoint therefore does not by itself establish that the transition from the current state is executable.

Physical feasibility and wearable plausibility are likewise different requirements. Robot reachability constrains PB-NBV's hemisphere, while Project Aria supplies calibrated device trajectories recorded during natural activities [Eng+23, Jia+25]. Robot-specific reachability can define a hard admissibility test for that platform; it cannot establish a distribution of realistic human head and body motion. For egocentric data generation, that distribution is an empirical prior to be measured, reported, and validated.

These distinctions map the proposed poses $q_{t,i} \in \mathcal{Q}_t$ to the canonical admissible row-index set $\mathcal{A}(s_t) = \{i : m_{t,i} = 1\}$, where $m_{t,i}$ is the hard-valid indicator after endpoint and transition checks. The utility or value function ranks only those admitted rows; a rejected candidate is unavailable rather than merely low-value [Jia+25, Str+24b]. Separating proposal, admission, and ranking makes the state dependence of available actions explicit and prepares the sequential decision model introduced next.

2.4 Sequential Decision-Making under Partial Observability

An egocentric reconstruction system never observes the complete physical scene. Its decision can depend on the sequence of earlier actions and observations, not only on the latest frame. In a partially observable Markov decision process, a belief state is a sufficient statistic of that action–observation history for policy choice [LHP23]. ARIA-NBV does not attempt general belief inference; the relevant consequence is narrower: any finite representation used for view selection is a hypothesis about which parts of the history are sufficient for predicting future target improvement.

The preceding section defined the admissible choices as the state-dependent row-index set $\mathcal{A}(s_t)$ after proposal and feasibility checks. That set can change as the target-conditioned reconstruction and current pose change, while a hierarchical controller can also condition camera position on an earlier look-at decision [Jia+25, Lu+26]. Sequential value must therefore be conditioned on the currently available action rather than treating every camera pose as permanently selectable.

Sequentiality matters when the best immediate view is not the best first step of a sequence. For target e, candidate i, and requested horizon h, the finite return and corresponding conditional value are

$$G_{t,e}^{(h)} = \sum_{k=0}^{h-1} \gamma^k r_{t+k}^e, \quad 1 \le h \le b_t \le H_{\max}$$

$$Q_{h,e}^{\star}(s_t, i) = \sup_{\pi \in \Pi^{\mathrm{act}}} \mathbb{E}_\pi\left[G_{t,e}^{(h)} \mid s_t, a_t = i\right], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t \le H_{\max}, \quad Q_{0,e}^{\star}(s, i) = 0$$

where continuation policies choose only from the admissible candidates at each step. Fixed-Horizon TD defines horizon-indexed predictions with the boundary $Q_0 = 0$ and relates horizon h to the shorter horizon $h - 1$ [De +20]. If an episode ends before all h rewards are collected, later rewards are zero. The value thus states the consequence of taking a candidate now and then following a named continuation rule; it is not an intrinsic property of the camera pose alone.

The horizon is distinct from two related bounds. The requested horizon h states how many future acquisition rewards the query represents; the remaining budget b_t limits how many acquisitions are still possible; and $H_{\max}$ marks the largest horizon supported by the chosen model and evidence. Universal Value Function Approximators establish the general possibility of conditioning one approximator on an additional task variable, while Fixed-Horizon TD shares representations across horizon-indexed predictions [De +20, Sch+15]. Neither result licenses extrapolation beyond the horizons actually supported and evaluated.

Learning such values from a fixed data collection introduces an epistemic constraint in addition to the mathematical horizon. Batch-constrained and conservative offline reinforcement-learning methods are motivated by the error that arises when a learned policy assigns high value to actions outside the data distribution [FMP19, Kum+20]. These methods differ in how they control that error, but they share the warning relevant here: a longer horizon does not create evidence for transitions that the data never resolves.

Together, partial observability, state-dependent choices, and finite-horizon return make candidate value explicitly relational [De +20, LHP23]. A camera pose has no context-free value. Its value depends on the causal information state, requested target, currently admissible support, horizon, state-update rule, and continuation policy. The next dependency is therefore representational. The actor state must preserve the distinctions on which those future consequences depend.

2.5 Egocentric and Geometric Representations

The preceding section established that future view value depends on a causal information state rather than on a camera pose alone. Project Aria provides calibrated, time-aligned egocentric streams, and EFM3D lifts posed image features and semi-dense geometry into a local gravity-aligned representation [Eng+23, Str+24a]. The representation question is therefore not how faithfully to reproduce the entire world, but which distinctions must survive so that target-specific counterfactual return remains predictable.

Four requirements follow. First, **causal sufficiency** requires the actor state to retain the observation history relevant to future return without importing future or unselected evidence. EFM3D supplies strong local evidence but has a finite voxel extent, while Hestia demonstrates that accumulated directional history can alter later choices [Lu+26, Str+24a]. A spatial state is therefore not sufficient merely because it is geometric; sufficiency is a hypothesis to be tested against the task.

Second, **spatial relationality** requires the target, observer, candidates, and observed support to be comparable in a common local geometry. Query-centric models express scene elements relative to the active query, reducing dependence on an arbitrary global origin [Zho+23]. For target-conditioned NBV, the relevant quantity is therefore not candidate position in isolation but candidate geometry relative to the current observer, target, and accumulated evidence. Coordinate choice also determines which changes should be treated as nuisances. Translating the world origin should not change a score, whereas gravity, metric scale, camera direction, target orientation, occlusion, and temporal order can remain informative [Bro+21]. The useful prior is thus relative geometry that removes arbitrary coordinates without erasing task-relevant physical structure.

Third, **candidate-order equivariance** follows from the action set itself. A candidate table denotes physical viewpoints rather than a ranked sequence, so permuting its rows must induce the same permutation of their scores; an invariant output would instead discard which score belongs to which action. Deep Sets provides invariant aggregation for shared set context, while Set Transformer provides permutation-equivariant candidate interaction [Lee+19, Zah+17]. This is a behavioral contract rather than an architecture prescription: independent row scoring, pooled context, and attention can all satisfy it.

Fourth, **physical observability** requires the representation to expose missingness and sensing geometry rather than treating unobserved space as negative evidence. Point models retain irregular surface samples and fine local relations, but their neighborhoods inherit the sampling pattern. Sparse-voxel models supply a regular convolutional structure while avoiding dense-grid cost, but introduce a chosen resolution and bounded spatial support. Equivariant message-passing models encode transformation rules directly, reducing the burden of learning them while committing the model to a chosen symmetry family [CGS19, SHW21, Zha+21]. These families therefore trade spatial fidelity, computational structure, and strength of geometric prior. None is an established improvement here until compared under the same observable inputs, target task, and endpoint utility.

These four requirements—causal sufficiency, spatial relationality, candidate-order equivariance, and physical observability—complete the conceptual dependency chain. The literature synthesis can now compare methods by the scientific distinctions they preserve, while the Method chapter remains responsible for one concrete realization and its tests [Bro+21, Str+24a].

2.6 Related-Work Synthesis and Research Gap

The reviewed approaches differ along five coupled dimensions: utility, target scope, represented evidence, candidate support and admission, and decision horizon [Fra+25, GML22, Jia+25, JLD24]. Comparing only one dimension can obscure the scientific task: a sequential coverage controller does not answer the same question as a greedy reconstruction-quality scorer, and neither comparison is meaningful if the relevant view lies outside candidate support.

Family	Utility and target scope	Represented evidence	Support, feasibility, and decision
PB-NBV [Jia+25]	Projected frontier and occupied coverage; reconstruction object	Classified voxel clusters and compact projection proxies	Reachability- and camera-conditioned partial hemisphere; greedy score
SCONE / MACARONS [GML22, Gué+23]	Expected surface coverage; scene-wide	Occupancy, visibility, or online reconstruction state	Iterative candidate selection
GenNBV / Hestia [Che+24, Lu+26]	Coverage gain; reconstruction object	Observation history plus geometric or directional state	Continuous or hierarchical policy; collision handling
FisherRF [JLD24]	Information gain; radiance-field scene model	Model-parameter uncertainty	Candidate views; greedy score
VIN-NBV [Fra+25]	Direct reconstruction improvement; reconstructed object	Candidate-conditioned projections of current reconstruction	Sampled candidates; greedy score
Next Best Sense [Str+24b]	Color–depth information gain; radiance-field scene model	3DGS uncertainty and multimodal observations	Feasible candidates; fallback after kinematic or planning failure
Object-centric 3DGS [Jeo+26]	Object-conditioned information gain; requested object	Per-Gaussian geometry, appearance, and object confidence	Candidate views; greedy score

Table 2: Concept-centred comparison of the literature families that bound the thesis question. Rows are distinguished by utility, target scope, represented evidence, candidate support and feasibility, and decision structure; they are not ranked by reported performance across incompatible settings.

The comparison exposes three tensions. The **objective tension** is that coverage and information criteria reward proxies for reconstruction, whereas VIN-NBV evaluates realized reconstruction error [Fra+25, GML22, JLD24]. The **temporal tension** is that sequential policies represent history and delayed reward, but the reviewed examples optimize coverage rather than the reconstruction quality of a requested target [Che+24, Lu+26]. Direct quality, target conditioning, and sequentiality are thus established separately under different assumptions.

The **support/state tension** is that a sequential objective is meaningful only relative to the actions that can be proposed and the causal information retained after acting. PB-NBV constrains proposal support, Hestia changes the state through directional history and endpoint repair, and Next Best Sense separates feasible candidates from execution fallback [Jia+25, Lu+26, Str+24b]. Even a perfect scorer cannot exploit a view outside its support, and a longer horizon cannot recover a distinction erased from the state.

This thesis evaluates the bounded conjunction left by those tensions: whether a finite-candidate policy can improve endpoint reconstruction quality for a specified target by valuing more than the next observation, using an egocentric geometric information state [De +20, Fra+25, Str+24a]. No reviewed work establishes that conjunction, and the literature cannot establish that the available actor state is sufficient or that non-myopic headroom exists in this setting. Those remain the empirical questions of the thesis.

The resulting principle is relational rather than architectural: view value is defined by a target, a causal state, admissible support, a horizon, a state update, and a continuation rule. The thesis does not generalize this bounded relation to exhaustive novelty priority, continuous control, actor-visible target discovery, closed-loop motion execution, or natural human movement [Che+24, Eng+23, Lu+26, Str+24b]. Chapter 3 now makes the bounded relation operational by separating observable state from the privileged geometry used to construct tasks and evaluate outcomes.

3 Oracle and Data Generation

Chapter 2 established that view value is relational: it depends on the target, causal state, admissible support, horizon, update rule, and continuation policy. This chapter turns that relation into a controlled experimental world. It first separates logged, selected-causal, and privileged information; then constructs target tasks and finite actions; and finally defines the intervention outcome and the immutable evidence needed to audit it. By the end, every value query has an explicit information boundary, action support, transition meaning, and outcome definition for the Method chapter to inherit.

3.1 Information Boundary

The learned component emits raw finite-horizon predictions for materialized candidates; separate masks govern selection and value supervision. Pre-forward inputs are protocol-admitted, not automatically deployable: `v0_gt_input` supplies a ground-truth-derived target control, whereas `v1_observed` requires observation-derived provenance. The experiment thus separates logged evidence, selected-action evidence, privileged control input, and offline counterfactual labels. ASE contributes both sensor-like streams and ground truth; EFM3D transforms only the logged subset into local 3D evidence [Met25, Str+24a]. Co-location in one replay row never makes those layers equally observable.

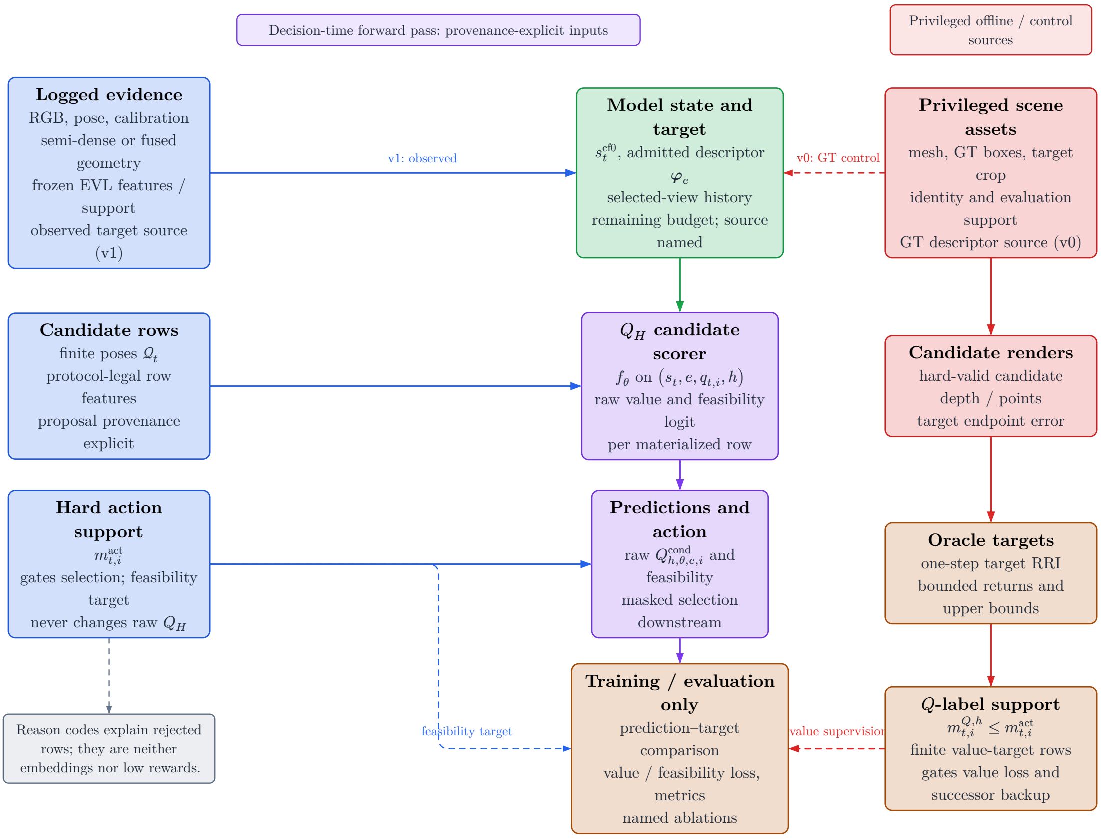

Figure 1: Actor and oracle boundary. `v0` admits privileged GT target input; `v1` requires an actor-visible descriptor. Raw row predictions are mask-independent: $m_{t,i}^{\text{act}}$ gates selection and feasibility supervision, while nested $m_{t,i}^{Q,h}$ support gates value loss and successor backup. Oracle renders and value targets enter only downstream.

The visibility boundary is protocol-relative and temporal. A dense render for an unselected candidate at the current decision step is oracle evidence. A render from an already selected action may enter a later state only under an explicitly named source protocol: privileged mesh-rendered depth, a declared sensor simulation, or an actor-visible observation. The same array shape can therefore denote different information, and source role must remain explicit.

3.1.1 Logged Observations

The target is external task context, so the value is conceptually $Q(s_t, e, a)$ rather than a target-independent state value. The logged history contains the recorded trajectory evidence,

$$s_t^{\mathrm{hist}} = \big(\boldsymbol{I}^{\mathrm{rgb}}, \boldsymbol{X}, \mathcal{P}^{\mathrm{semi}}, \boldsymbol{F}_v, e_t, b_t\big) \in \mathcal{S}^{\mathrm{hist}}$$

Calibrated image streams, trajectory, camera models, and gravity establish the spatial, metric, and temporal frame in which evidence is compared. EFM3D lifts posed image features into a finite gravity-aligned voxel volume rather than a complete world model [Str+24a]. An out-of-extent target or candidate therefore has missing representation support, not a measured zero and not automatically an invalid pose.

Semi-dense points add sparse surface evidence with uncertainty, timestamps, and observation lineage. EFM3D uses their observing camera centers to distinguish surface support from sampled free-space rays [Met25, Str+24a]. That distinction is causal: a missing point is not evidence that a voxel is free, and free-space evidence is justified only when its camera lineage is retained.

The target and remaining budget are decision context rather than sensing modalities. The current oracle task derives its target from a selected GT box; an actor-visible target must instead come from a declared observation-derived association path. The information role is fixed by provenance, not by tensor shape.

3.1.2 Selected Causal Evidence

The conceptual counterfactual actor state retains the immutable logged substrate and adds only information causally available after selected actions,

$$s_t^{\mathrm{cf0}} = \big(\boldsymbol{F}_v^{\mathrm{root}}, \mathcal{P}_t, \mathcal{Q}_t, m_{t,i}, \rho_{t,i}, e_t, b_t\big) \in \mathcal{S}^{\mathrm{cf0}}$$

For the canonical model, this compact tuple is read together with the explicit ordered selected-view history:

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\big(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\big)$$

In s_t^{cf0}, the root field remains fixed while the accumulated point set may grow only through selected observations. Candidate rows contain poses and proposal provenance, not observations from those poses. The action mask defines admission, the ordered history retains the factual approach sequence, and invalid-reason codes remain audit evidence rather than model input.

The current baseline materializes only root evidence and selected-pose history (the implementation anchor is `qh_cf0_v1`), a deliberately weaker `S0-pose` state:

$$s_t^{\text{S0-pose}} = \left(\mathcal{S}_{\text{root}}^{\text{VIN}}, \boldsymbol{F}_v^{\text{root}}, \left(\boldsymbol{T}_{r,e}, \boldsymbol{l}_e\right), \mathcal{Q}_t, m_{t,i}, \boldsymbol{H}_t^{\text{pose}}, b_t\right)$$

It carries root evidence, the admitted target descriptor, current candidates and hard mask, factual selected-pose prefix, and remaining budget. A richer state would update causal geometry after each selected action:

$$s_t^{\text{cf}+} = \left(s_t^{\text{cf0}}, \left(\boldsymbol{D}, \boldsymbol{V}, \mathcal{P}^{\text{cf}}, \boldsymbol{n}\right)_{1:t}\right)$$

The implemented richer control stops earlier: it persists selected mesh-rendered depth with calibration under the privileged `qh_cfplus_gt_depth_v1` source protocol. The carrier is causal with respect to the selected action but remains privileged and is not yet fused into surface, free-space, uncertainty, or a refreshed EFM3D field. It therefore cannot be interpreted as the full geometry-updated actor state.

The transition from t to $t+1$ may use only the observation produced by the selected row a_t. It may fuse its surface and ray evidence and update support, uncertainty, direction, and recency. It may not attach image features, detections, or EVL descriptors unless the protocol also provides the calibrated camera streams from which EFM3D derives them. Counterfactual geometry must therefore retain a source role and explicit absence masks for unavailable image-derived channels.

The counterfactual state is thus an **information state** for decision making, not automatically a complete Markov state of the physical scene. It becomes task-sufficient only if its retained root context, causal dynamic evidence, explicit history, target context, action support, and budget preserve every distinction needed to predict future target-specific return. `qh_cf0_v1` is the deliberately weaker `S0-pose` baseline; `qh_cfplus_gt_depth_v1` is a privileged depth carrier, not yet a realization of the geometry-updated state.

3.1.3 Privileged Counterfactual Evidence

The oracle state adds complete or counterfactual quantities outside the actor input graph:

$$s_t^{\mathrm{oracle}} = \left(s_t^{\mathrm{cf+}}, \mathcal{M}^{\mathrm{GT}}, \mathcal{M}_e^{\mathrm{GT}}, \boldsymbol{D}_q, \mathcal{P}_q, \mathrm{RRI}\right) \in \mathcal{S}^{\mathrm{oracle}}$$

$$\mathcal{M}_{\mathrm{NBV}} = \left(\mathcal{S}^{\mathrm{hist}}, \mathcal{S}^{\mathrm{cf0}}, \mathcal{S}^{\mathrm{oracle}}, \{\mathcal{A}_t\}, T, r_t^e, \gamma, H\right)$$

ASE provides per-frame metric GT depth aligned with RGB, per-pixel instance identifiers, class mappings, and the scene trajectory; the EFM3D release adds ASE OBB metadata and validation meshes for object-detection and surface-reconstruction supervision [Met25, Str+24a]. EFM3D uses the depth channel to supervise occupancy at sampled free, surface, and behind-surface points, while visible OBBs supervise centerness, class, and box geometry [Str+24a]. These uses establish label provenance, not actor observability.

For ARIA-NBV, $\mathcal{M}^{\mathrm{GT}}$ is the privileged reference surface used for candidate rendering and reconstruction evaluation, while $\mathcal{M}_e^{\mathrm{GT}}$ fixes the selected entity's evaluation support. All-candidate depth $\boldsymbol{D}_q$ and backprojected point sets $\mathcal{P}_q$ answer counterfactual queries for unselected rows. RRI then compares target reconstruction error before and after adding that candidate evidence. Depth, points, crops, and RRI are legal for label generation, oracle search, diagnostics, and named upper bounds; none is a contemporaneous input to the student value model.

A GT OBB may define the current oracle identity, pose, extent, crop, and target-bearing instruction. It does not show that the actor detected or associated that object. A deployable protocol must obtain the target hypothesis from EVL predictions or another observation-derived path, retaining ASE identity, segmentation, OBBs, and meshes only for matching and evaluation.

The information lattice therefore separates **what was logged**, **what a selected history causally adds**, and **what only the oracle can know**. Its layers are connected by controlled projections, not by freely copying arrays between stores. Persisting an oracle tensor beside actor evidence for efficient replay never changes its information role.

3.1.4 End-to-End Actor Visibility

Visibility answers whether a modality contains evidence for a spatial element from a particular view; feasibility answers whether an action is admissible; utility answers how beneficial an admissible action is for the target. These concepts must not be collapsed. A target can be weakly visible from a geometrically valid candidate, a candidate can lie outside EVL support while remaining physically reachable, and an oracle render can fail even though the candidate pose itself is feasible.

Invalidity is a constraint rather than a reward value. Geometry-invalid candidates receive a hard action mask and reason code before policy selection, stochastic normalization, and feasibility supervision. Value learning uses a separate label mask nested within action support: selected-row loss and next-state backup both require label support. A feasible candidate may still have low or negative target gain. Oracle-evaluation failure creates no candidate reason code; the configured recipe instead skips the row or table, or clears oracle-label validity and Q-training eligibility while reporting the failure separately. Oracle-derived feasibility must remain distinct from deployable validity when moving to an actor-visible target protocol.

Actor visibility is consequently an end-to-end property of the decision protocol. It includes how the target instruction is obtained, how candidate support is proposed, how the hard mask is computed, which selected observation updates the next state, and which fields reach the scorer. A scorer can be free of privileged tensors while still choosing from support oriented or pruned with privileged geometry. Such a protocol is a valid bounded oracle or control experiment, but its result cannot silently become a claim about independently actor-constructed actions.

3.2 Task and Action Construction

Validation TODO: Validate repeatability, failure handling, construct interpretation, root normalization, and the endpoint estimand before describing RRI as a reliable study objective.
Source: docs/literature/tex-src/arXiv-VIN-NBV; metric-validation artifacts
Gate: frozen metric protocol, repeatability table, and claim-level source locators

An oracle target task fixes entity identity, evaluation support, and the target instruction used to construct target-bearing actions. An actor-visible task would instead obtain its descriptor from observations. The present descriptor therefore defines a controlled requested object and its geometry; it does not establish actor-visible target discovery.

The general descriptor notation remains

$$\boldsymbol{\varphi}_e = \mathrm{Enc}_{\mathrm{tgt}}\Big(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\mathrm{proj}}, n_e^{\mathrm{semi}}, n_e^{\mathrm{EVL}}, \omega_e^{\mathrm{EVL}}, \ell_e^{\mathrm{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\Big)$$

The finite action interface consists of a candidate table $\mathcal{Q}_t$, hard validity mask $\boldsymbol{m}_t$, and invalid-reason vector $\boldsymbol{\rho}_t$. The admissible set is $\mathcal{A}_t = \big\{i : m_{t,i} = 1\big\}$. Invalid rows remain logged for coverage and failure analysis but lie outside policy argmax, sampling, loss targets, and bootstrap maximization. Low RRI is a valid low-utility outcome; it is never an encoding for infeasibility.

For target e, the oracle computes the target-cropped point–mesh error defined by the frozen outcome protocol. Candidate selection and Q_H supervision use root-normalized target gain; state-relative RRI is retained as a diagnostic. Fixed-budget endpoint gain is the intended policy estimand, but the current replay store records cumulative selected-chain gains rather than an independent post-horizon endpoint reconstruction for every policy. Confirmatory policy comparisons must therefore add matched oracle endpoint re-evaluation or an explicitly persisted endpoint record.

3.2.1 Target Selection

The implemented sampler does not match actor proposals to GT objects. It enumerates the non-padding GT OBB rows in a snippet, accepts rows with finite positive geometry, and applies seeded uniform sampling without replacement up to the configured per-snippet cap. Thus the stored `matched` status currently means geometry-valid GT task row, not successful proposal-to-identity association. IoU, ambiguity-gap, visibility, and support thresholds are absent from this admission rule.

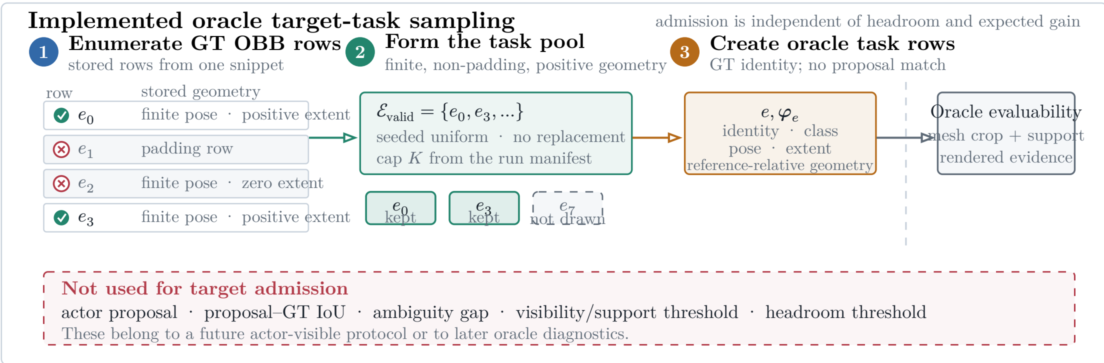

Figure 2: Implemented oracle target-task sampler. Geometry-valid GT OBB rows form the task pool, and seeded uniform sampling without replacement applies the manifest-defined cap. Rollout scoring later decides whether the selected crop is evaluable. No actor proposal, IoU match, visibility gate, or support gate is used.

The sampler bounds rollout cost without pre-filtering on headroom. Candidate scoring may subsequently invalidate the task when its mesh crop, current support, or rendered evidence is unusable; otherwise near-solved and negative-gain targets remain scientifically informative. A later actor-visible protocol must introduce proposal identity and observation-quality diagnostics as a separate selection stage rather than retroactively interpreting the present oracle fields as measurements.

3.2.2 Observed-Target Admission

The observed-target audit (implementation anchor `V1`) keeps the oracle task sampler separate from actor-visible target selection. For each observed or predicted target record $\hat{e}$, it compares only same-class GT OBB rows using oriented 3D IoU:

$$\mu_{\text{IoU}}(\hat{e}, e) = \text{IoU}_{3\text{D}}\left(\hat{\boldsymbol{B}}_{\hat{e}}, \boldsymbol{B}_e\right)$$

$$\tau_{\text{IoU}} = 0.20$$

A GT row qualifies only when its IoU is strictly greater than 0.20. The matcher admits the observed target iff exactly one GT row qualifies:

$$n_{\text{qual}}(\hat{e}) = \text{card}(\{e \in \mathcal{E} : \kappa(\hat{y}_{\hat{e}}, y_e) = 1 \wedge \mu_{\text{IoU}}(\hat{e}, e) > \tau_{\text{IoU}}\})$$

$$a_{\mathrm{id}}(\hat{e}) = 1 \,\mathrm{iff}\, n_{\mathrm{qual}}(\hat{e}) = 1$$

Equality at 0.20 is rejected, as are zero or multiple qualifying rows. The matching rule has no runner-up-gap or composite score. The current rollout sampler remains a distinct oracle-task path: geometry-valid GT-row admission is not evidence of actor-visible matching.

3.2.3 Candidate View Generation

Candidate generation turns one oracle instruction into a finite action table. Every quantitative choice—source population, target cap, shell size, family weights, motion limits, pruning thresholds, rollout recipes, renderer settings, and retention policy—belongs to the resolved run manifest and report bundle. This chapter defines semantics only; it does not designate one mutable configuration profile as canonical.

At rollout step t, candidate generation constructs a full shell. Here $\mathcal{K}_t$ is the resolved set of mixture components and n_k is the row count allocated to component k:

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad N_q = \sum_{k \in \mathcal{K}_t} n_k$$

with a fixed provenance component $k(i)$ per row. The core mixture contains forward-local, target-bearing, and lateral-bypass motion; the diversity challenger may additionally allocate mass to local refinement and revisit/backtrack components. The resolved manifest, rather than prose, determines which components and counts apply to a run.

For each row, the raw direction is sampled in the reference rig frame. The realistic profile uses the forward-biased Power Spherical distribution from the current implementation [DAT20]:

$$\boldsymbol{u}_i \sim \mathrm{PS}(\boldsymbol{e}_z, \kappa), \quad p(\boldsymbol{u}) = c_\kappa (1 + \boldsymbol{e}_z^T \boldsymbol{u})^\kappa, \quad \kappa = 8$$

The draw is mapped into configured azimuth and elevation caps without rejection. With $\psi = \mathrm{atan2}(u_x, u_z)$ and $u_y = \sin\theta$, the cap transform is

$$\psi' = \psi\frac{\Delta_\psi}{2\pi}, \quad y' = \sin\theta_{\min} + \frac{u_y+1}{2}\cdot(\sin\theta_{\max} - \sin\theta_{\min})$$

$$\boldsymbol{d}_i^0 = \mathrm{normalize}\Big(\Big(\sqrt{1-y'^2}\sin\psi', y', \sqrt{1-y'^2}\cos\psi'\Big)\Big)$$

The sampler draws a capped direction in the reference rig frame and reinterprets it as egocentric forward motion, target-bearing motion, lateral bypass, local refinement, or backtracking according to component provenance. Position and view provenance remain distinct: a component first constructs a base camera frame, then applies bounded local yaw and pitch jitter. In the core mixture, forward-local rows use the reference-rig base frame, whereas target-bearing and lateral-bypass rows use a target-look-at base frame. The supplied point is privileged in this chapter; calling the generator target-conditioned does not make it actor-visible.

The next figure separates two questions that become illegible when overlaid: where the decision is grounded in the 3D scene, and what support the complete finite table retains.

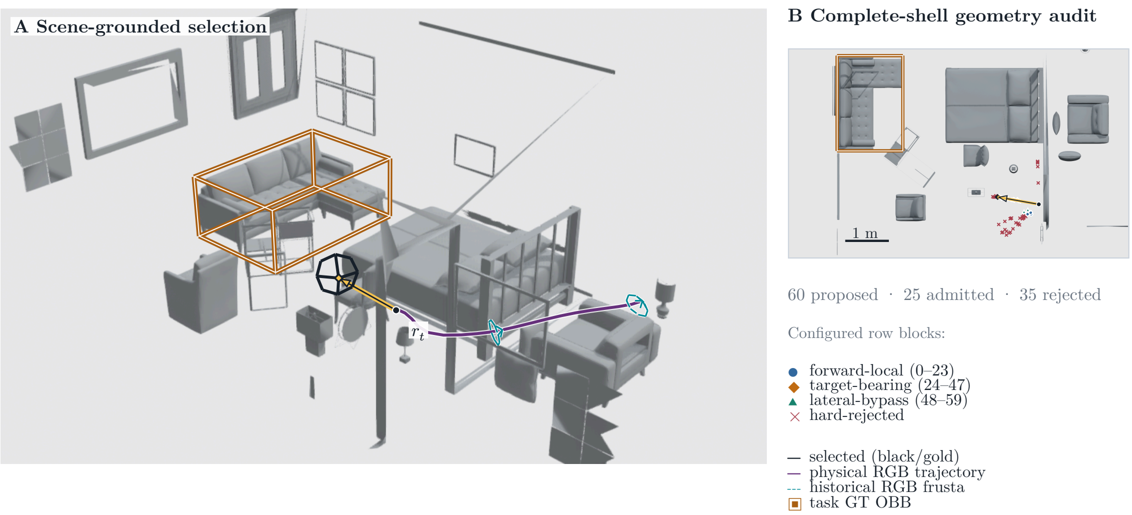

Figure 3: Scene grounding and support audit for one pinned ASE decision state. Panel A separates physical RGB history, the canonical sampling root, the selected oracle-task GT OBB, and the selected candidate view. Recovery reconstructs this pose from the preserved projection. Panel B retains all 60 stored centers and validity decisions: 25 admissible, 35 hard-rejected, and shell 47 selected. Shapes reconstruct configured shell blocks (24/24/12); stored row-level family identities were unavailable to this recovery export. Frusta approximate the calibrated Fisheye624 valid domain; the example fixes support and validity, not policy performance.

The three core position families then reinterpret this capped direction. The shared normalization operator is

$$\operatorname{normalize}(\boldsymbol{v}) = \frac{\boldsymbol{v}}{\|\boldsymbol{v}\|_2}, \quad \boldsymbol{v} \neq \mathbf{0}$$

Let $\boldsymbol{f} = \boldsymbol{e}_z$ be the rig-forward unit vector, $\boldsymbol{b}_e$ the supplied target bearing in the reference frame, $\boldsymbol{l}_e = \operatorname{normalize}\big(\boldsymbol{e}_y \times \boldsymbol{b}_e\big)$ the horizontal lateral direction, and $\boldsymbol{e}_y$ the world-up direction expressed in the sampling frame. The family directions are:

$$\boldsymbol{d}_i^{\text{forward}} = \operatorname{normalize}\big(\boldsymbol{f} + \alpha_f(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{f})\boldsymbol{f})\big), \quad \alpha_f = 0.45$$

$$\boldsymbol{d}_i^{\text{target}} = \operatorname{normalize}(\boldsymbol{b}_e + \alpha_t(\boldsymbol{d}_i^0 - (\boldsymbol{d}_i^0 \cdot \boldsymbol{b}_e)\boldsymbol{b}_e)), \quad \alpha_t = 0.4$$

$$\boldsymbol{d}_i^{\text{bypass}} = \operatorname{normalize}\Big(0.55\boldsymbol{b}_e + 0.85\,\operatorname{sign}\big(d_{i,x}^0\big)\boldsymbol{l}_e + \operatorname{clip}\Big(d_{i,y}^0, -0.35, 0.35\Big)\boldsymbol{e}_y\Big)$$

Finally, the sampler draws a component-resolved radius and transforms the reference-frame offset into world coordinates:

$$r_i \sim \mathcal{U}\Big(r_{\min,k(i)}, r_{\max,k(i)}\Big), \quad \boldsymbol{c}_i^w = \boldsymbol{T}_r^w\Big(r_i \boldsymbol{d}_i^{k(i)}\Big)$$

The forward-local family uses the reference-rig rotation as its base frame. Target-looking families instead construct a base frame from the supplied target center $\boldsymbol{p}_e$:

$$\boldsymbol{z}_i^w = \operatorname{normalize}(\boldsymbol{p}_e - \boldsymbol{c}_i^w), \quad \boldsymbol{y}_i^w = \operatorname{normalize}\big(\boldsymbol{e}_y - \big(\boldsymbol{e}_y^T \boldsymbol{z}_i^w\big)\boldsymbol{z}_i^w\big), \quad \boldsymbol{x}_i^w = \boldsymbol{y}_i^w \times \boldsymbol{z}_i^w$$

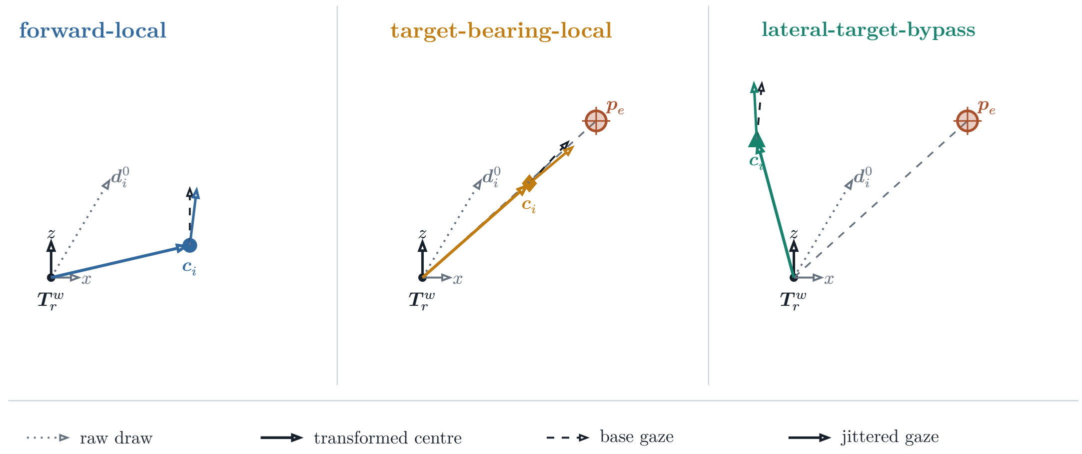

Figure 4: Geometric roles of the three core proposal families. Each panel applies one position transform to the same fixed raw direction; it then separates the component's base gaze from one bounded local-jitter realization. The plan view omits the bounded vertical term retained by the adjacent family equation. The construction explains proposal semantics only: radius sampling, family counts, feasibility, and selection remain separate.

These equations describe the sampler, not an optimal proposal distribution. Figure 4 isolates the two decisions that prose otherwise conflates: where a row proposes to move, and how its camera is oriented there. The forward-local transform preserves egocentric continuity, the target-bearing-local transform biases the proposed direction around the supplied target bearing, and the lateral-target-bypass transform introduces side-step views. Optional challenger families test smaller corrections and reversals. Candidate-profile utility must still be judged after pruning. A store in which target-aware families rarely survive cannot support a target-conditioned planning claim.

The optional `target_orbit` family is a bilateral partial-orbit endpoint proposal prior: it samples paired signed angular offsets around the supplied target while preserving the current horizontal target standoff. Feasibility pruning may remove either side, so bilateral attempt does not imply bilateral surviving support. This family is not part of the production mixture and is not a learned human-motion prior; it is an opt-in challenger for testing whether local orbit coverage improves candidate support.

Pruning converts the full shell into a compact valid-action table. A row remains valid only if it lies in the snippet occupancy support, stays clear of the GT mesh, avoids straight-line path collision, and satisfies local egocentric motion limits:

$$\|\boldsymbol{o}_i\|_2 \leq 1.0 \text{ m}, \quad |\Delta h_i| \leq 0.25 \text{ m}, \quad \max\big(0, -o_{i,z}\big) \leq 0.25 \text{ m}, \quad \Delta\psi_i \leq 70 \text{ deg}$$

The full shell is retained with position, strategy, mixture, sampling probability, rule masks, diagnostics, and invalid-reason bitsets. Panel B of Figure 3 makes the distinction concrete for one stored table: invalid rows remain inspectable but sit outside the admissible set. Invalid candidates cannot enter Q_H selection, stochastic normalization, or loss targets. Conversely, a feasible row with weak target support or low expected gain remains a valid low-utility example rather than receiving an invalid-reason code. A manifest-defined root-support threshold may reject an entire rollout task when too few actions remain:

$$N_{\text{valid}} \geq \max\big(12, \text{ceil}\big(0.25 N_q\big)\big)$$

This threshold is a data-support guard, not a reward threshold. Preflight reporting must retain rejected-root counts and failure reasons by candidate family.

3.2.4 Candidate-Support and Geometric-Coverage Diagnostics

Candidate quality is audited before reward interpretation. Let $\mathcal{I}_s$ be all attempted, non-padding rows in state s, $\mathcal{V}_s \subset \mathcal{I}_s$ the rows that pass actor-valid geometry and hard-action checks, $\mathcal{L}_s \subset \mathcal{V}_s$ the rows with a finite oracle label. The horizon-specific mask $m^{Q,h}_{s,i}$ selects rows in $\mathcal{L}_s$ for Q-training; it does not redefine the canonical candidate table $\mathcal{Q}_t$. The first transition is represented by the canonical action mask $m^{\text{act}}_{t,i}$; finite factual value-target availability is represented separately by $m^{Q,h}_{t,i}$. These are distinct populations: a row can be actor-valid without being renderable, oracle-labelled, or retained for training. The candidate actor-valid fraction is therefore

$$f_{\text{actor}(s)} = \frac{\sum_{i \in \mathcal{I}_s} m^{\text{act}}_{s,i}}{|\mathcal{I}_s|}$$

and hard valid support is

$$n_{\text{valid}(s)} = |\{i \in \mathcal{I}_s : m_i^{\text{act}} = 1\}|$$

Both quantities are computed per decision state before pooling. The configured family zero rate retains every configured family/state pair in its denominator, including pairs with zero valid rows:

$$z_{\text{family}(s)} = \left(\frac{1}{|}\mathcal{F}_s|\right) \sum_{f \in \mathcal{F}_s} \mathbb{1}\left[n_{\text{valid}(s,f)} = 0\right]$$

Every state statistic $q(s)$ is then averaged within its scene and finally macro-averaged across scenes:

$$\mathcal{S}_{c,q} = \{s \in \mathcal{S}_c : q(s) \ \text{ is defined}\}, \quad |(q)_c = \left(\frac{1}{|}\mathcal{S}_{c,q}|\right) \sum_{s \in \mathcal{S}_{c,q}} q(s), \quad \mathcal{C}_q = \left\{c \in \mathcal{C} : |\mathcal{S}_{c,q}| > 0\right\}, \quad |(q) = \left(\frac{1}{|}\mathcal{C}_q|\right) \sum_{c \in \mathcal{C}_q} |(q)$$

Here $\mathcal{S}_{c,q}$ is the metric-specific eligible-state set and $\mathcal{C}_q$ contains scenes with at least one eligible state. A scene with no defined state for q is absent from the descriptive scalar and counted in the accompanying undefined-scene table. A paired profile contrast uses the intersection of the two profiles' eligible-scene sets and reports every excluded scene. Thus the available-case denominator is explicit rather than an implicit row-level complete-case analysis.

The analysis manifest freezes the missing-state rule before result inspection. For support counts and fractions, a failed generation root contributes zero support; for the family-zero rate it contributes one. Target-relative balance and span are not numerically imputed when no target-conditioned direction is defined: the failed state remains in the failure table, and a paired scene is ineligible for a geometric profile contrast. Thus no missing state disappears through an implicit complete-case denominator.

For geometric comparison, candidate centres and target centres use the existing proposal-support normalization

$$d_{t,e}^{\text{current}} = \left\|\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w\right\|_2, \quad \tilde{\boldsymbol{c}}_{t,i}^{\text{support}} = \left(\boldsymbol{B}_{r,t}^{\text{Z-up}}\right)^\top \frac{\boldsymbol{c}_{t,i}^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\text{current}}}, \quad \tilde{\boldsymbol{p}}_{t,e}^{\text{support}} = \left(\boldsymbol{B}_{r,t}^{\text{Z-up}}\right)^\top \frac{\boldsymbol{p}_e^w - \boldsymbol{c}_{r_t}^w}{d_{t,e}^{\text{current}}}, \quad \left\|\tilde{\boldsymbol{p}}_{t,e}^{\text{support}}\right\|_2 = 1$$

: the origin is the factual expansion/root pose, the target has unit norm, the vertical axis is world-up, and distances are divided by the current root-to-target distance $d_{t,e}^{\text{current}}$. This is a target-relative support frame, not a camera image frame and not a claim that the root or target is visible. Side balance is computed from non-neutral attempted target-conditioned rows in the `target_bearing_local` and `target_orbit` families, using their target-relative azimuth signs. The neutral bin is $|y_{s,i}^{\text{target}}| \leq \varepsilon$ with $\varepsilon = 10^{-9}$ in normalized support coordinates; its count is reported but does not enter the two-sided balance denominator,

$$\ldots > \varepsilon\Big], \quad n_{-}(s) = \sum_{i \in \mathcal{J}_s^{\text{target}}} \mathbb{1}\Big[y_{s,i}^{\text{target}} < -\varepsilon\Big], \quad n_0(s) = \sum_{i \in \mathcal{J}_s^{\text{target}}} \mathbb{1}\Big[\Big|y_{s,i}^{\text{target}}\Big| \leq \varepsilon\Big], \quad b_{\text{side}(s)} = 1 - \frac{|n_{+}(s) - n_{-}(s)|}{n_{+}(s) + n_{-}(s \ldots}$$

and orbit coverage uses the circular minimum-covering span rather than a Cartesian maximum-minus-minimum range:

$$\alpha_{s,1} \leq \ldots \leq \alpha_{s,n_s}, \quad \alpha_{s,n_s+1} = \alpha_{s,1} + 2\pi, \quad s_{\text{orbit}(s)} = 2\pi - \max_{1 \leq j \leq n_s} \left(\alpha_{s,j+1} - \alpha_{s,j}\right)$$

The projection fraction records whether the calibrated target centre lies inside the calibrated image domain for evaluated rows in one state,

$$f_{\text{proj}(s)} = \frac{\sum_{i \in \mathcal{J}_s^{\text{evaluated}}} \mathbb{1}\left[i \in \mathcal{J}_s^{\text{projected}}\right]}{\left|\mathcal{J}_s^{\text{evaluated}}\right|}, \quad \left|\mathcal{J}_s^{\text{evaluated}}\right| > 0$$

and is then reduced through the same state–scene–cohort macro operator. A state with no evaluated projection rows is undefined and counted separately; it is not assigned a zero fraction or silently removed from a paired profile contrast. This remains an evaluation/framing diagnostic, not visibility or actor support. The finite-support oracle opportunity is the per-state maximum target-root gain,

$$o_{\text{oracle}(s)} = \max_{i \in \mathcal{L}_s} g_{s,i}^{\text{target-root}}, \quad |\mathcal{L}_s| > 0$$

which is defined only when the actor-valid, finite-label set $\mathcal{L}_s$ is nonempty. Its undefined-state count is reported, and eligible values are reduced through the same state–scene–cohort macro operator. The quantity measures available candidate headroom and is not a policy score. View jitter is likewise computed per state and macro-reduced. States without jitter rows, or without bounded rows for the cap-compliance diagnostic, are counted as undefined. The nonzero seminar-profile check remains separate from legacy zero-cap spherical support:

$$f_{\text{jitter}(s)} = \frac{\sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1] \cdot \mathbb{1}[|\Delta\psi_i| \le |(\psi)_i] \cdot \mathbb{1}[|\Delta\theta_i| \le |(\theta)_i]}{\sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1]}, \quad \sum_{i \in \mathcal{J}_s^{\text{jitter}}} \mathbb{1}[b_i = 1] > 0$$

The production mixture retains the seminar model's nonzero jitter invariant. Legacy `uniform_sphere` and `forward_powerspherical` rows with a zero cap are annotated as uncapped spherical support and are not represented by a bounded box.

Quantity	Population	Unit	Interpretation boundary
Actor-valid fraction	Attempted rows per state	fraction	Hard feasibility; not label or training availability.
Valid support	Actor-valid rows per state	rows	Report the lower tail and failed roots, not only a pooled mean.
Configured-family zero rate	Configured families per state	fraction	Retains absent and zero-support families; state values are scene-macro reduced.
Side balance / circular span	Non-neutral attempted target-conditioned rows	fraction / angle	Neutral rows and undefined states are reported separately; proposal coverage precedes pruning.
Target-centre projection	Evaluated candidate views per state	fraction	State–scene–cohort macro; zero-evaluated states are reported as undefined. Calibrated framing, not visibility.
Oracle opportunity	Finite actor-valid rewards per eligible state	gain	Undefined when no finite actor-valid label exists; available one-step headroom, not achieved policy return.
Jitter compliance	Bounded-jitter rows per state	fraction	State–scene–cohort macro; report undefined, nonzero, and uncapped support separately.

Table 3: Candidate-generation diagnostics and their aggregation populations. Every state-level quantity is reduced before scene and cohort aggregation.

3.2.5 Rollout Branch Sampling and Dataset Impact

Rollout recipes select and retain finite chains from the valid action table. The implemented families are uniform valid sampling, one-step oracle greedy selection, bounded oracle lookahead, and temperature-softmax sampling. Their horizon, branch factor, beam width, temperature, and seed are resolved parameters. These recipes generate replay diversity and bounded references; they do not constitute a learned policy.

The first action of the highest-ranked bounded-lookahead chain can differ from the action preferred by one-step greedy because lookahead ranks retained finite-horizon chains rather than immediate gain alone (Figure 5). The persisted artifact contains the beam-retained chains and their full per-step candidate shells; it is not an exhaustive materialization of the counterfactual action tree.

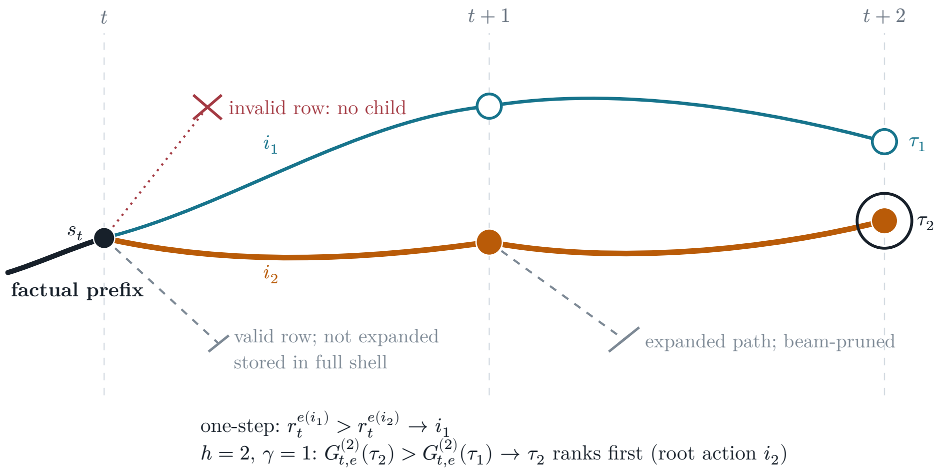

Figure 5: Constructed depth-two ordering reversal for target-root gain. Both solid paths are beam-retained from one factual prefix: i_1 has the larger immediate gain, while τ_2 ranks first at $h = 2$ and $\gamma = 1$; τ_1 also persists. The dotted stub is invalid. The single-bar stub is a legal shell row outside branch factor 2; the double-bar stub is an expanded path removed by beam width 2. Inequalities encode ordering only, not measured rewards or learned-policy performance.

For stochastic branches, let s_i be the finite oracle score of valid row i. The robust logit used for temperature-softmax is

$$\ell_i = \frac{s_i - \mathrm{median}(s)}{\mathrm{IQR}(s)\tau}, \quad P(i \mid m_i = 1) = \frac{\exp(\ell_i)}{\sum_{j:m_j=1}} \exp(\ell_j)$$

The target source defines the supervised task, the candidate mixture defines learnable action support, validity defines admissibility, and the recipe defines which chains enter replay. Reporting must therefore cover target-task coverage, valid fanout and invalid reasons by family, selected-family diversity, gain distributions, and retention cost before attributing policy behavior to non-myopic planning. The paired train-only pilots are bandwidth and candidate-profile probes; they are non-confirmatory and provide no held-out policy-performance evidence.

3.3 Measurement and Outcomes

Task construction and action admission specify the intervention; measurement specifies what consequence is compared. The oracle renders a selected candidate from the GT mesh, backprojects its depth, updates the privileged evaluation cloud through the frozen fusion rule, crops points and mesh to the selected target, and evaluates point–mesh error [Fra+25]. The crop, rendering, fusion, and scoring path are oracle-only.

3.3.1 Operational Reconstruction Error

Let $C_e(\mathcal{P}_t)$ denote the oracle-only crop of accumulated evaluation points to the selected target region. In the thesis protocol, its root points are reconstructed from the observed prefix of ASE GT depth; a candidate contributes renderer-derived depth that is backprojected and fused with the same root cloud. MPS semi-dense points remain actor-visible representation input and a legacy diagnostic source, not the current oracle evaluation cloud.

Writing $P = C_e(\mathcal{P}_t)$ and $M = \mathcal{M}_e^{\mathrm{GT}}$ for the target-cropped point set and mesh, the selected directional accuracy is the mean squared distance from each reconstructed point to its closest GT triangle,

$$D_{P\to M}\big(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}\big) = \frac{1}{\|\mathcal{P}\|} \sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

It penalizes reconstructed points that lie away from the target surface and therefore exposes extra, noisy, or misregistered geometry. The reverse term is the mean squared distance from each GT face to its closest reconstructed point,

$$D_{M\to P}\big(\mathcal{P}, \mathcal{M}^{\mathrm{GT}}\big) = \frac{1}{\|\mathcal{F}^{\mathrm{GT}}\|} \sum_{\boldsymbol{f}\in\mathcal{F}^{\mathrm{GT}}} \min_{\boldsymbol{p}\in\mathcal{P}} d(\boldsymbol{p}, \boldsymbol{f})^2$$

It penalizes target faces without nearby reconstructed evidence and therefore exposes missing support. Their sum is the bidirectional point–mesh error,

$$\Delta_t^e = d\big(C_e(\mathcal{P}_t), \mathcal{M}_e^{\mathrm{GT}}\big) = D^e_{P\to M,t} + D^e_{M\to P,t}$$

whose directional terms and total have units of square metres. The current and candidate-augmented clouds pass through the same deterministic fusion and point-cap configuration before scoring. This fusion step makes the before/after comparison operationally reproducible; it does not turn the point set into a continuous surface measure.

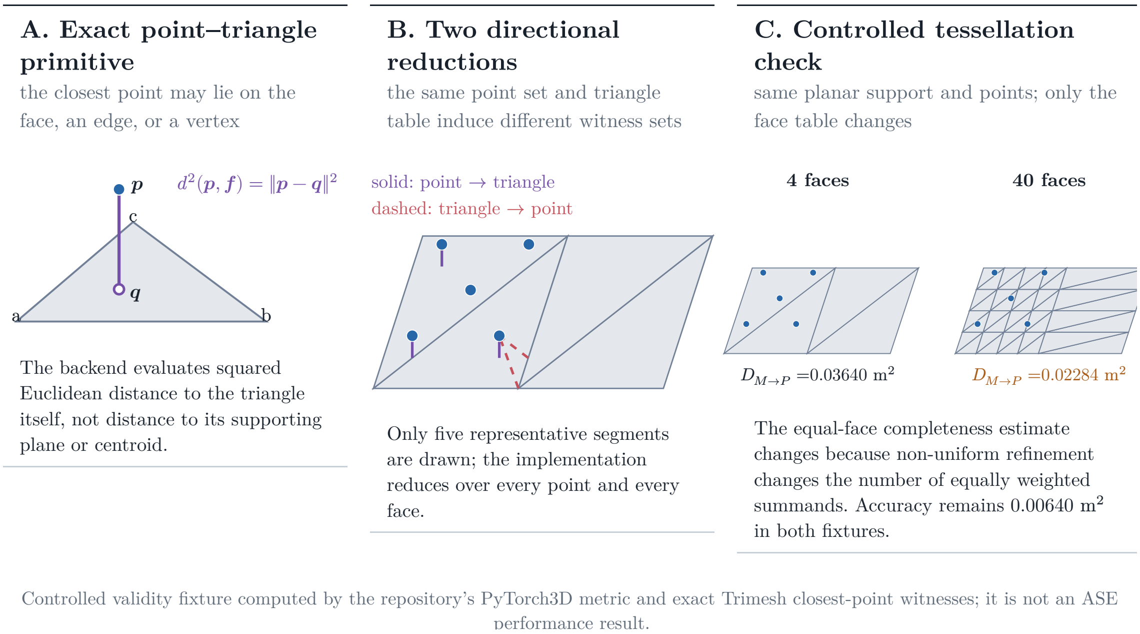

Figure 6: Point–mesh metric definition and controlled validity fixture. Panel A isolates the exact point-to-triangle primitive. Panel B distinguishes the point-to-face accuracy and face-to-point completeness reductions using computed closest-point witnesses. Panel C holds the planar support and point set fixed while changing only the triangle table; the measured equal-face completeness term changes from 0.03640 to 0.02284 m^2. These values are generated by the repository's PyTorch3D metric on a synthetic fixture and demonstrate a tessellation-sensitivity mechanism; they are not an ASE performance result.

The completeness reduction gives every mesh face equal weight. It is deliberately neither an area-weighted surface integral nor invariant to retessellation: subdividing part of an unchanged surface changes how much that region contributes. The controlled fixture in Figure 6 demonstrates this property. It bounds the interpretation of the operational estimand but does not invalidate comparisons that share one fixed mesh and scoring protocol.

3.3.2 Reliability under the Frozen Protocol

Reliability asks whether repeated execution of the same declared intervention produces stable errors and candidate rankings. It therefore covers rendering, backprojection, crop construction, point fusion and capping, numerical failure handling, and any stochastic sampling used by the metric. Deterministic code paths and a fixed mesh make the protocol reproducible in principle, but they do not establish repeatability across the intended study population. That evidence remains a prerequisite for RQ1 rather than an inferred property of the formula.

Metric family	**Question answered**	**Dependency and thesis role**
Equal-face bidirectional point–mesh error	Are reconstructed points close to GT faces, and does every GT face have nearby point evidence?	Depends on point fusion and mesh tessellation; selected operational RRI error.
Point-sampled Chamfer / accuracy–completeness	Are two sampled surface point sets mutually close?	Depends on sampling density, randomness, norm, and squaring convention; important alternative, not the current metric.
Thresholded precision, recall, and F-score	What fraction of predicted and reference samples fall within tolerance τ, and how do the two fractions balance?	Interpretable at a physical tolerance but threshold-dependent; useful diagnostic rather than the scalar RRI owner.
TSDF, occupancy, coverage, or information gain	How much volumetric surface, free space, or previously unknown space is reconstructed or observed?	Depends on grid, truncation, and visibility definitions; answers a different question from target surface quality.

Table 4: Reconstruction-metric alternatives and their role in this thesis. No alternative is promoted to a co-primary metric.

For comparison, a common squared point-sampled Chamfer convention draws a reference set Q_e from the mesh and averages nearest-neighbour distances in both directions,

$$D_{\text{PS-Chamfer}(\mathcal{P},\mathcal{Q})} = \frac{1}{\|\mathcal{P}\|} \sum_{\boldsymbol{p}\in\mathcal{P}} \min_{\boldsymbol{q}\in\mathcal{Q}} \|\boldsymbol{p}-\boldsymbol{q}\|_2^2 + \frac{1}{\|\mathcal{Q}\|} \sum_{\boldsymbol{q}\in\mathcal{Q}} \min_{\boldsymbol{p}\in\mathcal{P}} \|\boldsymbol{q}-\boldsymbol{p}\|_2^2$$

Its apparent symmetry does not remove sampling-density dependence: changing the number or distribution of samples changes the finite approximation. Some benchmarks also use unsquared distances, so the norm and squaring convention are part of the metric definition rather than an interchangeable naming detail. At tolerance τ, precision counts reconstructed samples with a reference neighbour closer than τ, recall counts reference samples with a reconstructed neighbour closer than τ, and $F_{1,\tau} = 2\text{Prec}_\tau \frac{\text{Rec}_\tau}{\text{Prec}_\tau + \text{Rec}_\tau}$. These scores are physically interpretable but can change with the selected threshold and discard error magnitude on either side of it. Volumetric coverage and information gain are valuable for exploration and mapping, yet they measure discovered space or uncertainty reduction rather than the target reconstruction-quality change defined here.

3.3.3 Construct Validity

Construct validity asks what the operational error represents. In this thesis it measures idealized target-surface evidence under ASE depth, a selected target crop, equal-face mesh reduction, and the declared reconstruction update. It is not a representation-independent measure of object quality, perceived quality, or scene completeness. The rendering and fusion operator are part of the estimand: two systems using the same point–mesh formula but different state updates do not measure the same intervention effect.

The implementation crops mesh faces when any vertex lies inside the oriented target Oriented Bounding Box (OBB) and evaluates the selected point–mesh scorer. Geometry-invalid candidates are removed by the hard action mask and retain persisted candidate reason codes. Empty mesh crops, insufficient current support, or unusable renders instead invalidate the separate oracle evaluation: the recipe either skips the affected row or table, or clears oracle-label validity and Q-training eligibility, while reporting the oracle failure independently rather than assigning a candidate reason code.

Every compared policy must therefore share the same target mesh and crop, ASE-depth source, renderer and backprojection, fusion and point cap, and metric configuration. Replacing ASE depth with MPS semi-dense evidence would define a different intervention and would require separate analysis of texture, gradient, uncertainty, and sampling effects.

3.3.4 RRI, Training Reward, and Endpoint Gain

The thesis objective is not the absolute point–mesh error alone, but its normalized reduction after a fixed acquisition budget. It distinguishes state-relative RRI, the additive target-root reward, and endpoint gain: all are dimensionless reductions of the same target-cropped error, but their denominators and scientific roles differ.

The state-relative marginal target RRI remains the VIN-compatible candidate diagnostic:

$$\mathrm{RRI}^e_{t,i} = \frac{\Delta^e_t - \Delta^e_{t|i}}{\max(\Delta^e_t, \varepsilon)}$$

The implementation also reports its selected-chain running sum:

$$C^{\mathrm{RRI},e}_t = \sum_{k=0}^{t-1} \mathrm{RRI}^e_{k,a_k}$$

Because the marginal RRI denominator changes with state, this cumulative diagnostic does not telescope to endpoint improvement. The immediate training reward instead adapts the same error reduction to a target crop and normalizes by the rollout-root target error [Fra+25]:

$$r^e_t = \frac{\Delta^e_t - \Delta^e_{t+1}}{\max(\Delta^e_0, \varepsilon)}$$

With matched sequential states and unit discount, its selected-chain sum is the root-normalized endpoint difference under that same clamped root denominator:

$$J^e_t = \sum_{k=0}^{t-1} r^e_k = \frac{\Delta^e_0 - \Delta^e_t}{\max(\Delta^e_0, \varepsilon)}$$

The finite-horizon return is the discounted sum of those target rewards along a selected counterfactual branch. It is a training target for Q_H, not a claim that the deployed system has an online continuous-control policy:

$$G^{(h)}_{t,e} = \sum_{k=0}^{h-1} \gamma^k r^e_{t+k}, \quad 1 \le h \le b_t \le H_{\max}$$

Endpoint gain is the primary fixed-budget comparison metric because it measures the target quality after the same number of acquisitions for each policy:

$$J^{(H)}_e = \frac{\Delta^e_0 - \Delta^e_H}{\Delta^e_0 + \varepsilon}$$

The log-gain variant is retained only as an internal scale-sensitivity ablation:

$$J^{(H)}_{e,\log} = \log(\Delta^e_0 + \varepsilon) - \log(\Delta^e_H + \varepsilon)$$

$J_e^{(H)}$ is the fixed-budget evaluation metric, $G_t^{(H)}$ is the learning return, and $J_{e,\log}^{(H)}$ is a sensitivity diagnostic. The endpoint equation uses $\Delta_0^e + \varepsilon$, whereas the additive target-root reward uses $\max(\Delta_0^e, \varepsilon)$; the two are therefore not canonically interchangeable, even when geometry, horizon, and acquisition budget match. State-relative one-step RRI remains a VIN-compatible diagnostic. The resolved manifest must freeze the discount, clipping, target cap, crop policy, and all evaluation-geometry parameters for each reported experiment.

3.4 Evidence and Reproducibility

The experiment preserves two evidence roles. An immutable source store owns the logged actor substrate and one-step oracle products; a replay store references those rows and owns target tasks, retained chains, per-step candidate shells, hard masks, reason codes, and selected-action successor evidence. This boundary prevents counterfactual experiments from mutating their source and prevents dense one-step labels from being mistaken for factual multi-step transitions.

Lineage follows the scientific units. One source may define several target tasks; one task may produce several recipe-specific retained chains; one chain contains ordered selected steps; and each step owns a full finite candidate shell. The store therefore represents selected or beam-retained evidence rather than an exhaustive counterfactual tree. Derived padded training arrays remain caches over these factual relations. Invalid rows stay auditable while action, training, and bootstrap masks exclude them from their respective decisions.

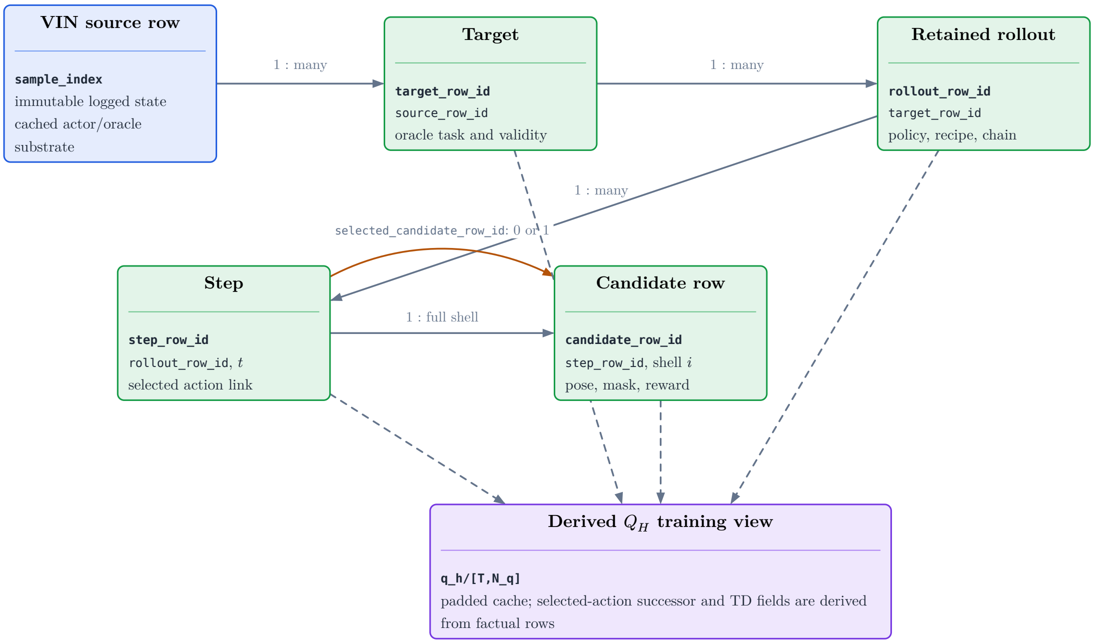

Figure 7: Normalized lineage of the replay evidence. An immutable source row may define several target tasks; each target may produce several retained policy chains; each chain contains ordered steps; and each step owns one full candidate shell. Selected-action successor and finite-horizon training fields are derived from those factual relations rather than treated as an independent counterfactual transition table.

Evidence family	Scientific interpretation
Sources and targets	Manifest-backed task coverage; not proof of actor-visible target discovery.
Candidates and invalidity	Full-shell support with separate hard-action, training, padding, and future deployable-feasibility roles.
Retained chains and steps	Recipe-selected evidence; not a persisted exhaustive search tree.
Selected depth	Chosen-action successor observation with calibration and source role; actor input only under an explicitly admitted later-state protocol.
Finite-horizon training view	Derived training cache whose rewards and masks must agree with factual rows; not a scene-memory representation.
Candidate-support metrics	Attempted-row actor-valid fraction, per-state valid support, configured-family zero rate, attempted target-conditioned side balance and circular span, calibrated target-centre projection fraction, finite-support oracle opportunity, and bounded-jitter compliance.
Metric populations	State metrics aggregate state then scene then cohort; failed roots and zero-valid configured family/state pairs remain in denominators. Projection is framing, oracle opportunity is headroom, and jitter is QC—not visibility or policy performance.

Table 5: Scientific interpretation of replay evidence. Numeric values are rendered from the resolved report bundle in the experiment and reproducibility sections.

Selected-depth persistence stores only the depth raster and calibration for the chosen action at each retained step. It can reconstruct the selected-observation prefix without duplicating dense all-candidate renders, but persistence does not make the source actor-visible. A privileged reader may consume selected mesh-rendered depth; an actor-visible reader must consume an admitted sensor-like or observed source. Unselected candidate renders remain oracle-only, and selected depth is not an independently scored endpoint artifact.

Likewise, rollout rows summarize final cumulative selected-chain metrics; they do not preserve every rejected branch or a policy-neutral endpoint reconstruction. These limitations must be resolved by matched endpoint re-evaluation before confirmatory policy comparison.

Missingness is part of the evidence rather than an ordinary zero. Reporting therefore retains target-task coverage, candidate validity and failure reasons, family survival and selection, selected-history sanity, gain distributions, source-role counts, and runtime or storage exclusions. Exact schema columns, joins, compression, chunking, hashes, and execution commands belong to the reproducibility record and resolved manifests. Development bandwidth pilots may size later jobs but cannot support held-out reconstruction or policy claims.

4 Method

Chapter 3 fixed the experimental world: a target-specific task, a finite and hard-masked action set, a causal replay transition, and a root-normalized reconstruction-gain outcome. This chapter fixes the learner that operates inside that world. The selected method is a scalar-horizon, target-conditioned candidate-value model. It reads root evidence, relative candidate and target geometry, the factual selected-pose prefix, and remaining budget; it predicts a conditional value for every materialized candidate before the authoritative hard mask is applied.

The evaluated configuration uses the `S0-pose` state, H0 mean-pooled pose history, A1 candidate-to-state cross-attention, and direct continuous Huber regression. A0, an independent-row MLP with the same inputs and decoder, is the matched interaction control. Both execute the same value task; their held-out comparison remains pending. Richer selected-depth state, ordered history, ordinal decoding, and candidate-set interaction remain alternatives because none has yet passed a frozen comparative evaluation.

The value query distinguishes factual remaining budget b_t from requested residual horizon h. Dense one-step labels anchor Q_1; factual selected transitions provide the only admissible recursive path to Q_h for $h > 1$. Exact horizon two is consequently the first epistemic test of learned lookahead: it checks whether recursion recovers a target that can be computed without trusting a learned longer-horizon continuation. Passing that test is necessary but not sufficient for a policy claim, which additionally requires positive oracle headroom and held-out endpoint recovery.

The chapter proceeds from the selected state and encoding, through finite action and replay semantics, to architectural acceptance properties and the finite-horizon learning objective. This order keeps failures interpretable: lost state information, malformed action or replay semantics, and value-learning error remain distinct explanations rather than an undifferentiated model defect.

4.1 Selected Actor State

Implementation: implemented **Evidence: pending**

The selected method uses the `S0-pose` actor state: immutable root evidence, an admitted target descriptor, the current finite candidate table and hard mask, factual selected-pose history, and remaining budget. The interface is implemented and tested; task sufficiency and policy value remain unestablished.

Literature: [Str+24a]

Promotion gate: name the actor-state and target-source protocols in every run; establish held-out task sufficiency

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/tests/lightning/test_qh_module.py

The actor state is the information available before selecting action a_t, not every quantity co-located in a replay row. Oracle target gains, mesh crops, ground-truth associations, current all-candidate renders, and labels are excluded from the scorer graph. The non-deployable `v0_gt_input` target protocol supplies privileged target geometry for a controlled oracle task; `v1_observed` requires a descriptor constructed from observations and binds its source and construction provenance. These protocols define different claims even when their tensors share a shape.

The selected state read is

$$\boldsymbol{h}_{t,e,i} = \mathrm{Read}\big(\boldsymbol{\Phi}_t^{\mathrm{scene}}, \boldsymbol{h}_e^{\mathrm{tgt}}, q_{t,i}, \boldsymbol{H}_t, t, H\big)$$

Its immutable root component contains semidense evidence and lossy global moments of locally supported EFM3D features. Its dynamic component is only the strictly causal selected-pose prefix. Candidate regeneration updates the reference pose, prefix, remaining budget, and action table; it does not imply a new RGB observation, a refreshed EFM3D field, or fused selected depth.

State component	Selected evidence	Interpretation boundary
Root scene	semidense and supported EFM3D summary	Immutable logged context; missing local support is not measured free space.
Target	protocol-admitted identity and geometry	Privileged oracle instruction or actor-visible descriptor, never an unlabelled mixture.
Dynamic history	selected poses $\boldsymbol{H}_t$	Causal action history, but no selected appearance, depth fusion, or free/unknown memory.
Decision context	candidate rows, hard mask, b_t, and h	Finite support and value query; invalidity remains external to conditional value.

Table 6: Selected `S0-pose` state protocol. The table states what the method consumes and, equally importantly, what it cannot represent.

4.1.1 Sufficiency and the local-evidence limit

A decision state is task-sufficient only if it preserves every distinction that can change target-specific future return. `S0-pose` cannot satisfy that condition by construction when two histories share poses but reveal different surfaces, occlusions, or free-space evidence. It is therefore a deliberately compact baseline and an interface test, not a claim that pose history is a complete reconstruction state.

EFM3D reinforces this limit. Its gravity-aligned voxel field has an explicit pose and finite extent [Str+24a]. A target or candidate outside that extent may remain physically valid while lacking local lifted evidence. Global moments further remove spatial correspondence. The selected method can test whether this compact context contains usable signal, but a failure cannot be attributed to value learning alone, and a success cannot establish spatial state sufficiency. Richer causal surface or ray memories are kept outside the selected method until a frozen comparison isolates their added information.

4.2 Target, Candidate, and History Encoding

Implementation: implemented **Evidence: pending**

The selected encoder materializes one target-conditioned query per candidate from root evidence, local relative geometry, H0 selected-pose history, remaining budget, and requested horizon. Its tensor contract is tested; scientific usefulness remains pending.

Literature: [Bro+21, Sch+15]

Promotion gate: preserve local-frame, mask, source, and strictly causal prefix tests; evaluate held-out ranking

Development source: aria_nbv/aria_nbv/data_handling/qh_data/views.py; aria_nbv/aria_nbv/vin/modules/pooling.py; aria_nbv/aria_nbv/vin/modules/qh_history_encoders.py; aria_nbv/tests/vin/test_qh_history_encoders.py

For target e and candidate $q_{t,i}$, the scorer input is the structured relation

$$\mathcal{I}_{t,e} = \left(\boldsymbol{h}_e^{\text{tgt}}, \boldsymbol{\Phi}_t^{\text{scene}}, \boldsymbol{H}_t, \boldsymbol{b}_t, t, H, \left\{\boldsymbol{x}_{t,i}, \boldsymbol{e}_{a|i}^{\text{rel}}\right\}_{i=1}^{N_q}\right)$$

rather than a flat replay row. Source and availability masks distinguish absent evidence from measured zeros; padding is separate from physical invalidity; supervision and audit lineage never enter the learned query.

4.2.1 Target-conditioned query

The target descriptor separates identity, geometry, observed support, confidence, and source:

$$\boldsymbol{\varphi}_e = \text{Enc}_{\text{tgt}}\left(\hat{\boldsymbol{B}}_e, \hat{\boldsymbol{y}}_e, \hat{\pi}_e, A_e^{\text{proj}}, n_e^{\text{semi}}, n_e^{\text{EVL}}, \omega_e^{\text{EVL}}, \ell_e^{\text{src}}, \boldsymbol{T}_{r_t,e}, \boldsymbol{T}_{c_t,e}\right)$$

The learned target token combines only fields admitted by the named target protocol with target-local scene support:

$$\boldsymbol{h}_e^{\text{tgt}} = \text{MLP}_{\text{tgt}}(\text{concat}(\boldsymbol{\varphi}_e, \boldsymbol{g}_e^{\text{tgt}}))$$

Current oracle tasks populate ground-truth-derived target geometry and leave some generic descriptor fields unmeasured. Availability must therefore be explicit. Replacing an absent measurement by an ordinary zero would make source provenance indistinguishable from scene evidence.

4.2.2 Relative candidate geometry

Canonical world poses remain reproducibility facts, but the model receives a candidate pose relative to the current decision frame,

$$\boldsymbol{T}^{\mathrm{rel}}_{r_t,i} = \boldsymbol{T}^{-1}_{w,r_t}\boldsymbol{T}_{w,c_{t,i}}, \quad \boldsymbol{\delta}^{p}_{r_t,i} = \boldsymbol{R}^{\top}_{r_t}\big(\boldsymbol{c}_{t,i} - \boldsymbol{c}_{r_t}\big), \quad \boldsymbol{R}_{r_t,i} = \boldsymbol{R}^{\top}_{r_t}\boldsymbol{R}_{t,i}$$

with continuous pose features

$$\boldsymbol{h}^{\mathrm{pose}}_{t,i}(q_{t,i}; r_t) = \mathrm{concat}\Big(\boldsymbol{\delta}^{p}_{r_t,i}, \mathrm{R6D}\big(\boldsymbol{R}_{r_t,i}\big), \big\|\boldsymbol{\delta}^{p}_{r_t,i}\big\|_2, \mathrm{atan2}\big(\delta^{y}_{r_t,i}, \delta^{x}_{r_t,i}\big), \Delta h_{t,i}, \boldsymbol{u}^{\mathrm{up/frustum}}_{t,i}\Big)$$

and an explicit candidate–target relation:

$$\boldsymbol{h}^{\mathrm{rel}}_{t,e|i}\big(q_{t,i}, e\big) = \mathrm{concat}\Big(\boldsymbol{\delta}^{p}_{e|i}, \big\|\boldsymbol{\delta}^{p}_{e|i}\big\|_2, \cos\theta^{\mathrm{opt}}_{t,e,i}, \beta^{\mathrm{elev}}_{t,e,i}, \lambda^{\mathrm{obb}}_{t,e,i}\Big)$$

This factorization removes arbitrary world origin and yaw conventions while retaining physical variables such as gravity, scale, height, target orientation, bearing, and camera direction. The candidate row is a geometric query, not a copy of the scene. Sampler family and generator provenance remain audit variables unless a named ablation admits them.

4.2.3 Strictly causal pose history

For every realized state, the reader supplies the complete selected-pose prefix $j < t$ expressed from the factual current camera. The selected H0 encoder applies the shared pose map and a padding-aware arithmetic mean:

$$\boldsymbol{p}^{\mathrm{hist}}_{t,j} = \mathrm{PoseEnc}\Big(T_{c_t \leftarrow c_j}\Big), \quad j < t$$

$$a^{\mathrm{hist}}_{t,j} = \frac{t - 1 - j}{H_{\max}}$$

$$\boldsymbol{h}^{\mathrm{hist}^{\mathrm{H0}}}_{t} = \mathrm{HistProj}\left(\frac{1}{\max(1,t)}\sum_{j<t}\boldsymbol{p}^{\mathrm{hist}}_{t,j}\right)$$

$$\boldsymbol{h}^{\mathrm{hist}^{\mathrm{H1}}}_{t} = \mathrm{HistProj}\Big(\mathrm{LastValid}\Big(\mathrm{CausalTransformer}\Big(\Big[\boldsymbol{e}_{\mathrm{empty}}, \big\{\boldsymbol{p}^{\mathrm{hist}}_{t,j} + g\big(a^{\mathrm{hist}}_{t,j}\big)\big\}_{j<t}\Big]\Big)\Big)\Big)$$

Only the H0 branch of this equation belongs to the selected method. The root state uses a learned empty token; padded entries cannot affect the history summary. Prefix cardinality and chronology are not substitutes for remaining budget b_t or requested horizon h, which remain separate tokens. Because H0 observes poses rather than selected surfaces or free-space evidence, it inherits the `S0-pose` sufficiency limitation identified above.

The maximum horizon $H_{\max}$ binds data, model, and checkpoint. Remaining budget records the factual state; h selects a requested return within $1 \le h \le b_t \le H_{\max}$. A syntactically valid query still fails closed if its horizon lacks manifest-bound training and evaluation support. The public interface scores one scalar horizon at a time and preserves candidate order in its $[B, S, N_q]$ output.

4.3 Finite Action and Causal Replay

Implementation: implemented **Evidence: pending**

Finite candidate tables, hard masks, factual selected transitions, and dense one-step label admission are implemented and tested. Population support and policy outcomes remain pending.

Promotion gate: retain deterministic row identity, source roles, hard-mask isolation, and selected-transition validation

Development source: aria_nbv/aria_nbv/rollouts/replay/engine.py; aria_nbv/aria_nbv/rollouts/zarr_store.py; aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/tests/rollouts/test_qh_reader.py

At step t, the generator returns the full candidate table $\mathcal{Q}_t$, a hard-valid mask $\boldsymbol{m}_t$, and versioned failure reasons. The admissible action set is

$$\mathcal{Q}_t = \{q_{t,i}\}_{i=1}^{N_q}, \quad \mathcal{A}_t = \{i \in \{1, ..., N_q\} : m_{t,i}^{\text{act}} = 1\}$$

Invalid rows remain auditable but cannot enter selection, Q supervision, or a bootstrap maximum. A feasible row may have negative utility; infeasibility is therefore never encoded as a small value. Padding, materialization, `valid_action_mask`, `q_train_mask`, and any predicted feasibility remain distinct. Under the dense-valid supervision profile, the writer proves and the reader revalidates equality of Q-label support and hard action support on every realized state. Unknown or legacy label-density profiles fail closed for this claim.

Candidate orientations factor a component-specific base gaze from bounded yaw–pitch perturbations. Paired proposal components may reuse a camera centre across two base-gaze families, but both remain separate actions. This makes translation and viewing direction distinguishable without collapsing their values. Generator family, sampled support, hard rejection, and stable shell identity remain attached to every row so proposal coverage can be audited before policy quality is interpreted.

4.3.1 Factual transition

After selecting an admitted row, the replay engine applies

$$(x_{t+1}, \boldsymbol{H}_{t+1}, b_{t+1}, \mathcal{Q}_{t+1}) = \mathrm{Step}\big(x_t, \boldsymbol{H}_t, b_t, q_{t,a_t}, \xi_t\big)$$

where x_t is the current reference pose, $\boldsymbol{H}_t$ the selected-pose prefix, b_t the remaining budget, and ξ_t the deterministic generation context. The next candidate table is regenerated around the selected pose under the same target task and versioned generator. Proposal and action-selection randomness use separate streams keyed by the factual selected-action history, so reordering retained trajectories cannot change a previously defined state.

This is the complete transition of the selected `S0-pose` method. It changes reference pose, selected-pose history, remaining budget, and finite action support. It does not claim that the actor received RGB, fused depth, or updated a spatial reconstruction. Selected mesh-rendered depth may be persisted as privileged counterfactual evidence for a separate control, but unselected candidate renders never enter a successor state.

The normalized replay view expands one retained chain into its realized decision states. Each state sees only the prefix available before its action. Several states from one chain may share a batch, but that tensor colocation does not create temporal communication. The factual selected action supplies the only successor link used by finite-horizon learning; no counterfactual transition is fabricated for unselected rows.

Training, validation, and test populations must share reward, mask, target protocol, actor-state protocol, horizon, and label-support semantics. They may use different candidate generators or behavior policies because those fields describe population shift rather than alter the value definition. Each split therefore binds its own support provenance, and acquisition remains scene-disjoint and independent of oracle outcomes or model error.

4.4 Selected Interaction and Acceptance Conditions

Implementation: implemented **Evidence: pending**

A1 candidate-to-state cross-attention is the selected interaction; A0 is its feature-matched independent-row control. Both pass the same executable acceptance tests. Their scientific comparison remains pending.

Literature: [Bro+21]

Promotion gate: retain row-equivariance, invalid-row isolation, frame, source, mask-independence, and horizon tests; measure the A0/A1 control

Development source: aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/vin/modules/qh_state_fusion.py; aria_nbv/tests/vin/test_qh_state_fusion.py; aria_nbv/tests/vin/test_target_finite_horizon.py

Each materialized candidate is an independent query over shared scene, target, history, budget, and horizon tokens. A1 uses the candidate as query and those five state tokens as keys and values; candidates never attend to other candidates. A0 flattens the same ordered tokens and applies a row-shared MLP. Both expose the same-width context to the same value-decoder seam.

Control	Interaction	Interpretive role
A0	independent-row MLP	Tests whether the frozen descriptors and objective are learnable without attention.
A1	candidate-to-state cross-attention	Selected model; tests whether query-dependent reading of shared state improves the same value task.

Table 7: Admitted interaction pair. Inputs, state protocol, decoder, objective, and hard-mask owner remain fixed; parameter count and runtime must still be reported.

Candidate order has no semantic meaning. Jointly permuting aligned rows by Π must permute predictions identically:

$$f_\theta\big(\Pi \boldsymbol{X}_t, \Pi \boldsymbol{m}_t^{\text{cand}}\big) = \Pi f_\theta\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\text{cand}}\big)$$

Changing only hard-invalid row contents must not alter valid-row predictions:

$$\operatorname{Mask}\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\text{cand}}\big) = \operatorname{Mask}\big(\boldsymbol{X}_t', \boldsymbol{m}_t^{\text{cand}}\big) \quad \Rightarrow \quad \operatorname{Mask}\big(f_\theta\big(\boldsymbol{X}_t, \boldsymbol{m}_t^{\text{cand}}\big), \boldsymbol{m}_t^{\text{cand}}\big) = \operatorname{Mask}\big(f_\theta\big(\boldsymbol{X}_t', \boldsymbol{m}_t^{\text{cand}}\big), \boldsymbol{m}_t^{\text{cand}}\big)$$

The scorer reads the materialization mask only to sanitize padding. It never reads the hard action mask; training owns label and bootstrap admission, and online inference owns the final selectable set. Thus changing the hard mask cannot change raw conditional values or feasibility logits. Duplicate-row and valid-count tests further prevent accidental set normalization from redefining the value of an unchanged physical candidate.

Local-frame encoding removes arbitrary global origin conventions without claiming exact SE(3) equivariance. Gravity, scale, height, yaw, camera direction, target orientation, and frustum geometry remain meaningful physical variables [Bro+21]. Provenance tests additionally exclude target gains, meshes, associations, and current candidate renders from the actor graph. Previously selected privileged depth belongs only to an explicit non-deployable state protocol.

Finally, $h = 1$ has no bootstrap, $h > b_t$ is rejected rather than clamped, and an $h > 1$ target may depend only on a factual successor queried at $h - 1$. No monotonicity across horizons is assumed because admissible actions may have negative immediate gain.

4.5 Target-Conditioned Finite-Horizon Value

Implementation: partial **Evidence: pending**

The selected method is A1–H0–S0 with a scalar requested horizon, direct continuous Huber regression, fitted-Q recursion, and hard-masked online selection. The implementation path is tested, but learned exact-Q2 accuracy and policy evidence remain absent.

Literature: [De +20, EGW05, HGS15, Sch+15]

Promotion gate: populate a frozen held-out exact-Q2 receipt; establish oracle headroom; evaluate endpoint recovery on untouched scenes

Development source: aria_nbv/aria_nbv/vin/models/target_finite_horizon.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/lightning/qh_q2_certification.py; aria_nbv/aria_nbv/oracle/pipelines/online_qh.py; aria_nbv/tests/lightning/test_qh_q2_certification.py

4.5.1 Conditional scorer

The actor input contains root semidense and supported EFM3D summaries, target geometry admitted by the declared source protocol, relative candidate geometry, the factual selected-pose prefix, remaining budget, and materialization support. Oracle gains, target crops, meshes, associations, and audit lineage are excluded.

The current `root_moments_v1` scene summary and H0 pose history make this an executable `S0-pose` baseline, not a task-sufficient reconstruction state.

Before the hard action mask is applied, the scorer emits

$$\left(Q^{\mathrm{cond}}_{h,\theta,e,i}, \ell^{\mathrm{feas}}_{t,i}\right) = f_\theta\left(s_t, e, q_{t,i}, h\right), \quad h = b_t \text{ in implemented V1}; \quad 1 \le b_t \le H_{\max}$$

for every materialized row. `conditional_q` alone carries return units and enters regression, Bellman backup, and online ranking. Feasibility logits are a separate auxiliary prediction; multiplying them by value would mix meanings. The training adapter owns hard-valid label and bootstrap admission, while online inference owns the final masked selection.

The selected A1 interaction lets each candidate query shared scene, target, history, budget, and horizon tokens. A0 supplies the matched independent-row control. Both use the same geometric query and value decoder. Their comparison therefore asks whether query-dependent state reading helps under a fixed state and objective; it does not isolate attention independently of parameter count or runtime.

4.5.2 Scalar requested horizon

For target e, the return over exactly h residual actions is

$$G^{(h)}_{t,e} = \sum_{k=0}^{h-1} \gamma^k r^e_{t+k}, \quad 1 \le h \le b_t \le H_{\max}$$

and the learned value is

$$Q^\star_{h,e}(s_t, i) = \sup_{\pi \in \Pi^{\mathrm{act}}} \mathbb{E}_\pi\left[G^{(h)}_{t,e} \mid s_t, a_t = i\right], \quad i \in \mathcal{A}_t, \quad 1 \le h \le b_t \le H_{\max}, \quad Q^\star_{0,e}(s,i) = 0$$

One model call follows the frozen interface

$$f_{\theta\left(s_t^{\text{S0-pose}}, \boldsymbol{\varphi}_e, \{q_{t,i}\}_{i=1}^{N_q}, h\right)} \to \left(\left\{Q^{\mathrm{cond}}_{h,\theta,e,i}\right\}_{i=1}^{N_q}, \left\{\ell^{\mathrm{feas}}_{t,i}\right\}_{i=1}^{N_q}\right)$$

The notation Q_H names the bounded family rather than a collection of separately configured models. Remaining budget b_t describes the factual state; h selects one return from the triangular domain $\{(b_t, h) : 1 \le h \le b_t \le H_{\max}\}$. Omitting h requests the full factual residual budget. The mathematical boundary $Q_0 = 0$

closes recursion, whereas zero in executable tensors denotes padding and is never a learned query.

Dense labels train $h = 1$ for every hard-valid candidate. For $h > 1$, only the factual selected action has a successor observation, so recursive supervision is necessarily narrower. A bundle records realized support by horizon and rejects any requested horizon that lacks frozen training and evaluation evidence. A wide syntactic interface is not evidence of wide learned capability.

4.5.3 Direct continuous objective

The decoder maps each candidate representation directly to root-normalized finite-horizon return. Huber loss retains the metric order and additive units of that outcome while limiting the leverage of extreme targets. Losses are first averaged within realized state and then within horizon so large candidate tables and abundant one-step labels do not silently dominate recursive rows. Reported calibration and support remain stratified by horizon.

At $h = 1$, evaluation can use dense candidate labels to report continuous error, within-state pairwise ranking, and greedy regret. Factual $h > 1$ transitions do not supply counterfactual values for unselected rows, so longer-horizon ranking and regret remain undefined until exact or independently oracle-rescored candidate tables exist. Recording zero support is more faithful than inventing a metric from the selected transition alone.

4.5.4 Fitted-Q recursion

Batch fitted Q iteration turns the fixed replay collection into successive supervised problems [EGW05]. The immediate reward is

$$r_t^e = \frac{\Delta_t^e - \Delta_{t+1}^e}{\max(\Delta_0^e, \varepsilon)}$$

and for $h > 1$ the selected method uses the lower-horizon target

$$y_t^{(h,e)} = r_t^e + \gamma B_t^{(h,e)}$$

where the continuation value is detached, frozen, or supplied by a delayed copy. The structural rule is $Q_h \leftarrow Q_{h-1}$: a horizon never bootstraps from itself. The successor maximum intersects the factual horizon support with the hard action set,

$$\mathcal{A}_{t+1}^{(Q,h-1)} = \left\{ i : m_{t+1,i}^{\text{act}} = 1 \wedge m_{t+1,i}^{Q,h-1} = 1 \right\}$$

and Double Q separates row selection from delayed evaluation [HGS15]. This estimator may limit max bias, but it cannot repair unsupported actions, aliased state, or missing selected observations.

4.5.5 Why exact horizon two is decisive

When the selected action has a successor table with dense one-step labels, its finite-support two-step target is computable exactly:

$$y_t^{(2,\text{exact})} = r_t^e + \gamma_t \max_{j:m_{t+1,j}^{\text{train}}=1} r_{t+1,j}^e$$

This makes horizon two the first non-myopic falsification test. A unit test can inject exact Q_1 values and prove that the tensor path implements the equation. A frozen population test instead measures learned recursion error,

$$\varepsilon_t^{(2)} = \left| y_t^{(2,\text{recursive})} - y_t^{(2,\text{exact})} \right|, \quad \varepsilon_t^{(2)} \le \tau_{\text{abs}} + \tau_{\text{rel}} \left| y_t^{(2,\text{exact})} \right|$$

on held-out supported rows. The latter tests the learned Q_1 approximation, state encoding, masks, successor linkage, and recursive target construction together. It is therefore model evidence rather than another implementation test.

The held-out evaluation begins with the complete eligible chain population rather than only rows on which an exact target is available. Chains that terminate or lose support before $h = 2$ remain in the denominator with zero exact rows. Scene- and target-stratified coverage, error, uncertainty, numeric tolerance, candidate width, generator, recipe, and behavior policy are frozen before inspection. Low error on a small supported subset cannot justify a longer horizon when minimum support or coverage fails.

Exact Q_2 is necessary but not sufficient: policy claims additionally require positive equal-budget oracle headroom and held-out endpoint recovery.

5 Experimental Design

The experiment is a sequence of admissibility gates rather than one aggregate model score. Measurement must be stable before a population can be interpreted; population and action support must exist before oracle headroom can be measured; headroom must be meaningful before learned recovery is relevant. The learned path then separates actor-visible one-step value, exact two-step recursion, and fixed-budget endpoint recovery. Each stage has its own population, estimate, uncertainty, and stopping rule.

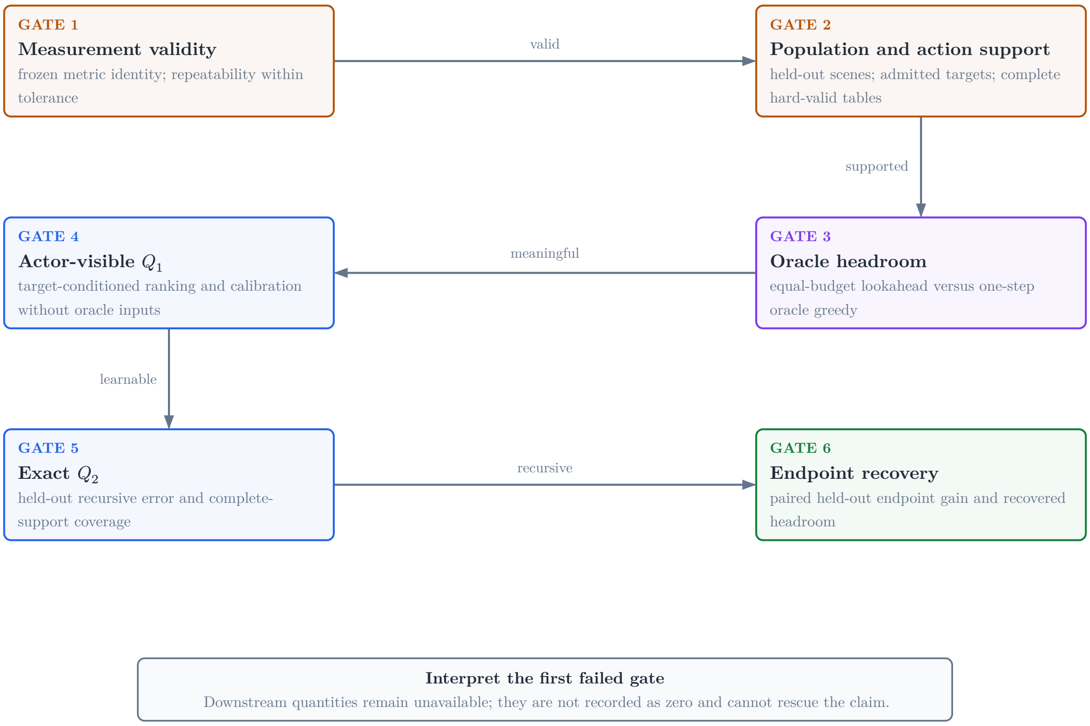

Figure 8: Evidence-gated claim path. The first failed gate determines the admissible interpretation; later quantities remain unavailable rather than being treated as zero or pooled into a composite score.

5.1 Population, Estimands, and Gate Order

Validation TODO: Preregister the eligible population, exclusions, scene aggregation, independent-run structure, uncertainty interval, meaningful headroom, recovery fraction, and comparison family before inspecting confirmatory outcomes.

Source: experiment manifest and analysis specification

Gate: immutable analysis plan plus matched held-out policy table

5.1.1 Candidate-Support Estimands and Aggregation

RQ4 treats the scene as the independent unit. Candidate-support quality-control observations are computed per state, averaged across states within scene, and finally macro-averaged across the scene cohort; no candidate row is treated as an independent scene replicate. These within-profile diagnostics remain descriptive. Any comparative claim instead uses a paired per-scene difference over eligible scene/target/root tasks observed under both profiles, equal candidate budgets, and the frozen candidate-generation manifest. Its report identity is `paired_scene_mean_difference`, with the paired scene count as its denominator and an interval computed over the paired scene differences.

Missing or failed states are retained in the audit tables. Support counts and fractions use the preregistered failure values stated in Table 3; an undefined target-relative direction prevents that paired scene from entering a geometric contrast rather than creating an angle by imputation. The excluded-scene count and reason remain reported. Thus the descriptive state–scene macro summaries and the paired comparative estimand have distinct populations and cannot be substituted for one another.

Minimum operational support is reported as a worst-case diagnostic; the fifth percentile of per-state valid support is the more stable lower-tail estimand, accompanied by actor-valid fraction, configured-family zero rate, target-side balance, and circular target-relative orbit span. Failed roots and zero-valid configured family/state pairs remain part of the denominator. The projection fraction, oracle opportunity, and jitter compliance are secondary diagnostics: projection is calibrated framing rather than visibility, oracle opportunity is finite-support headroom rather than policy performance, and jitter compliance must preserve the nonzero production seminar-jitter invariant.

The source population comprises ASE/ATEK snippet windows whose configured GT mesh and object-box table resolve. A frozen manifest assigns entire scenes to train, validation, or test before model selection; snippets from one scene cannot cross those boundaries. Every report records the manifest hash, scenes, snippets, admitted target tasks, rollout chains, realized transitions, candidate rows, exclusions, and failure strata. A capped training pilot may test throughput or support, but it cannot estimate held-out policy performance.

The current task generator samples geometry-valid ground-truth boxes. It can therefore establish oracle-task coverage, not actor-visible target discovery. Ground-truth target geometry may define task construction, labels, and bounded oracle references, but an actor-facing claim additionally requires an observation-derived descriptor and an audit of its matching and failure population. The same boundary excludes unselected candidate renders and oracle-derived labels from decision-time input.

The evidence sequence in Figure 8 answers the research questions in a fixed order:

1. **Measurement validity (RQ1):** freeze crop, render, fusion, and point–mesh metric identity; show repeatability within a declared tolerance.
2. **Population and action support (part of RQ4):** establish scene-disjoint target-task coverage, candidate-family survival, hard validity, and acquisition feasibility with exact denominators.
3. **Oracle headroom (first half of RQ2):** compare bounded lookahead with one-step oracle greedy under the same acquisition budget,

$$\Delta_{\text{look}} = J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{oracle-1}})$$

4. **Actor-visible Q_1 (RQ3 and RQ4):** evaluate target-conditioned one-step ranking, calibration, target matching, and dense-label replay coverage without privileged actor input.

5. **Exact Q_2 (RQ2 and RQ4):** measure held-out two-step error, factual-successor coverage, and horizon support against the finite-support target.
6. **Endpoint recovery (second half of RQ2):** only after meaningful headroom, estimate the prespecified recovered fraction

$$\eta_Q = \frac{J_e^{(H)}(\pi_Q) - J_e^{(H)}(\pi_{\text{learned-1}})}{J_e^{(H)}(\pi_{\text{oracle-look}}) - J_e^{(H)}(\pi_{\text{learned-1}}) + \varepsilon}$$

   from matched endpoint oracle evaluation.

RQ5 and RQ6 are evaluated only if the offline finite-candidate evidence justifies extending the action or interaction setting. This ordering prevents an attractive downstream policy estimate from compensating for an unstable metric, an unsupported action set, privileged actor input, or failed recursion.

5.2 Replay Admission for the Learning Gates

Implementation: partial **Evidence: pending**

Dense one-step supervision, selected-transition fitted-Q recursion, scalar horizon support, and exact-Q2 certification are executable. No qualifying held-out learning or policy evidence is currently available.

Literature: [De +20, EGW05, HGS15]

Promotion gate: held-out dense-Q1 checkpoint; qualifying exact-Q2 receipt; matched endpoint oracle evaluation

Development source: aria_nbv/aria_nbv/rollouts/qh_reader.py; aria_nbv/aria_nbv/lightning/qh_module.py; aria_nbv/aria_nbv/lightning/qh_q2_certification.py

Replay eligibility is defined by factual evidence, not padded tensor shape. The hard action mask identifies selectable rows. The stricter one-step training mask also requires a finite target-root-gain label. A recursive row additionally requires the factual selected action, its reward and terminal role, and—when nonterminal—a reproducible successor state with lower-horizon support. Modality presence, source role, target validity, action validity, Q-label support, transition support, and horizon support remain separate masks.

Learning gate	Estimand	Minimum factual evidence
actor-visible Q_1	immediate root-normalized target gain	hard-valid candidate, finite one-step label, actor-visible target protocol, and held-out scene role
exact Q_2	first recursive finite-support target	selected reward plus terminal outcome or a successor table with complete one-step labels over every hard-valid row
recursive Q_h	greedy value within represented support	factual selected transition, terminal and discount; nonterminal rows also require successor state, hard mask, and admitted Q_{h-1} support

Table 8: Learning-target admission. Dense one-step labels, exact two-step targets, and general recursive targets are different evidence populations.

The exact-Q_2 completeness rule is strict. One finite successor label is insufficient because an unlabelled hard-valid action could hold the maximum. The certification denominator therefore begins with every eligible held-out chain, retains chains that terminate or lose support, and reports how many reach a complete successor table and a factual exact row. Missing support fails the gate; it is not zero recursion error.

One retained rollout contributes several causal decision states. State t receives only the prefix observed before a_t, even if later states share the same batch. Dense Q_1 loss uses all admitted candidate rows; $h > 1$ loss uses only factual selected transitions. Candidate losses are averaged within state, then states within horizon, then non-empty horizon means uniformly. This keeps large action tables and abundant one-step rows from hiding recursive failure.

The learner implements fitted-Q recursion from Q_h to a detached or delayed Q_{h-1} target and uses the hard-valid successor set for every maximum [De +20, EGW05]. Double-Q selection and evaluation address max bias but do not create transition support [HGS15]. The experiment therefore reports horizon-specific population, calibration, terminal fraction, bootstrap support, and exact-Q_2 coverage before any endpoint comparison.

The promotion boundary is deliberately narrow. Executable training, a valid checkpoint bundle, or low error on a few rows cannot establish finite-horizon value. Actor-visible Q_1 must first pass on held-out scenes; exact Q_2 must then pass its frozen independent-unit, coverage, and tolerance rules; endpoint recovery is interpreted only if oracle headroom is already meaningful.

5.3 Matched Policies and Failure Attribution

Implementation: partial **Evidence: pending**

Scene-paired aggregation and reporting that preserves missingness are implemented; the confirmatory policy population is absent.

Promotion gate: frozen held-out manifest, completed paired endpoints, and validated confirmatory report bundle

Development source: aria_nbv/aria_nbv/rollouts/reporting.py; docs/typst/thesis/experiment_data.typ

The scene is the independent experimental unit. Comparisons are paired by scene, target task, root state, candidate distribution, hard-validity regime, and acquisition horizon. Repeated snippets, targets, and seeds are aggregated within scene before inference so dense sampling cannot masquerade as independent evidence. Policies receive the same number of acquisitions even when their planner depth differs.

Policy	Decision information	Budget	Scientific role
π_{rand}	actor-visible	H	valid-action lower reference
$\pi_{\text{learned-1}}$	actor-visible	H	learned myopic control
$\pi_{\text{oracle-1}}$	oracle immediate gain	H	one-step oracle-greedy comparator
$\pi_{\text{oracle-look}}$	bounded oracle lookahead	H	privileged bounded-lookahead reference
π_Q	actor-visible	H	finite-horizon recovery policy

Table 9: Matched policy roles. Oracle access defines a privileged reference, not a deployable upper bound; every row retains the same acquisition budget.

The primary estimand is the paired per-scene difference in fixed-budget endpoint target-quality gain. Crop, mesh, rendering, backprojection, fusion, point cap, and point–mesh metric identity are shared across policies. The analysis manifest freezes exclusions, within-scene aggregation, interval procedure, comparison family, meaningful headroom, and recovered-headroom threshold before learned-policy outcomes are inspected. Ratios are reported only when their headroom denominator passes the meaningful-effect gate.

First failed gate	Admitted interpretation	Not implied	Next discriminating evidence
measurement	the oracle outcome is not stable enough for comparison	anything about planning, learning, or support	repair and repeat the frozen metric protocol
population / action support	the requested estimand lacks an adequate study or action population	zero utility or policy failure	report exclusions, family survival, horizon coverage, and resource failures
oracle headroom	no meaningful non-myopic structure was detected in the frozen setup	universal myopia or model inadequacy	change support or horizon only in a separately declared study
actor-visible Q_1	the available actor information does not recover immediate target value	a specifically long-horizon failure	audit target matching, leakage, calibration, and state support
exact Q_2	the first learned recursion is unsupported or inaccurate	endpoint planning value or a need for more architecture	separate coverage, Q_1 error, successor linkage, and bootstrap error
endpoint recovery	the admitted learned policy does not recover prespecified headroom	which mechanism failed	stratify by support, target observability, replay coverage, and state aliasing

Table 10: Failure-attribution matrix. Interpretation stops at the first failed gate; downstream architecture stories remain hypotheses.

Validation TODO: Populate the six gates in order. Missing upstream evidence blocks downstream quantities rather than becoming a zero result.

Source: confirmatory report bundle and exact-Q2 receipt

Gate: artifact-backed Results chapter

Open decision: Freeze aggregation, interval level, comparison family, meaningful headroom, and recovery threshold in the resolved analysis manifest.

Source: confirmatory analysis plan

Gate: analysis freeze before outcome inspection

Train-only pilots and failed generation attempts remain feasibility evidence. Renderer out-of-memory events can motivate batching and resource measurement, but they cannot support or refute candidate utility, oracle headroom, or policy quality. Likewise, a policy that cannot continue remains in the paired analysis with no further gain rather than disappearing from the denominator.

6 Results

Validation TODO: Populate one result row per inferential stage from a confirmatory bundle: population, estimate, uncertainty, admitted conclusion, and blocking condition. Every value must resolve to its raw and derived artifact provenance.

Source: confirmatory report bundle, exact-Q2 receipt, and analysis manifest

Gate: all six evidence gates resolve without fixture or pilot substitution

The loaded report declares evidence class *pilot*. Schema validity proves that provenance and missingness are readable; it cannot promote a development fixture, training pilot, or incomplete store. Availability is therefore cumulative: a downstream gate is reported only when every prerequisite and its own required facts are present.

Gate / RQ	Population	Estimate	Uncertainty	Admitted conclusion	Blocking condition
measurement / RQ1	frozen repeated oracle evaluations	repeatability statistic: not available	declared numeric tolerance	metric comparison is admissible	mismatched identity, absent repeats, or tolerance failure
population/ action / RQ4	held-out scenes, targets, and full candidate tables	coverage and failures: not available	exact denominators and scene strata	the population relevant to oracle headroom is described	missing population, exclusions, or valid-action support
headroom / RQ2	paired lookahead and one-step oracle scenes	endpoint effect: not available	paired scene interval	bounded setup contains meaningful non-myopic structure	measurement/support failure or non-meaningful effect
actor Q_1 / RQ3	held-out actor-visible candidate states	ranking and calibration: not available	scene-clustered interval	actor information recovers immediate target value	privileged input, matching failure, or inadequate calibration
exact Q_2 / RQ2	eligible held-out factual successors	error and coverage: not available	independent-unit tolerance rule	first recursive target is recovered	incomplete support or failed recursion tolerance
endpoint recovery / RQ2	paired held-out learned and reference policies	recovered fraction: not available	paired scene interval	learned policy recovers prespecified headroom	any earlier gate or recovery-rule failure

Table 11: Evidence availability by inferential stage. “Not available” preserves missingness and names the blocker; it is not a zero estimate.

6.1 Measurement Validity

The loaded evidence does not establish repeatability of the frozen target-specific endpoint metric. All downstream policy quantities therefore remain unavailable.

6.2 Population and Action Support

No validated held-out bundle currently defines the study population and complete candidate-support denominators. Training-source reachability and renderer failures remain feasibility observations only.

6.3 Oracle Headroom

Meaningful equal-budget lookahead headroom over one-step oracle greedy is not estimable from the loaded evidence.

6.4 Actor-Visible One-Step Value

No validated held-out result currently shows target-conditioned one-step ranking and calibration from actor-visible inputs.

6.5 Exact Two-Step Recursion

No qualifying held-out exact-Q2 receipt is available; recursive finite-horizon accuracy is therefore unestablished.

6.6 Endpoint Recovery

The thesis cannot estimate learned endpoint recovery because the prerequisite gate chain is incomplete.

6.7 Resource Feasibility

Renderer memory failures motivate bounded rendering and retained failure provenance, but no validated completed-store evidence supports throughput or dataset-volume extrapolation.

7 Discussion

The available evidence supports an implementation conclusion, not a policy conclusion. ARIA-NBV represents target-specific finite actions, separates actor-visible inputs from oracle supervision, applies invalidity as a hard constraint, records factual selected transitions, and reports observed values without replacing missing evidence. These properties make the proposed study auditable. They do not establish metric repeatability, held-out population support, non-myopic headroom, learned value accuracy, or endpoint improvement.

The evidence chain in Figure 8 makes this distinction constructive. The first unresolved gate owns the current interpretation. Here, repeatability and the held-out population are not established, so headroom, actor-visible Q_1, exact Q_2, and endpoint recovery remain unavailable—not negative. Treating them as zeros would collapse missing evidence into measured failure and could make a pipeline limitation appear to answer the scientific question.

7.1 Interpreting future outcomes

The failure-attribution matrix in Table 10 separates mechanisms that an endpoint score alone cannot identify. Unstable measurement invalidates every downstream comparison. Adequate measurement but insufficient target or action support locates the problem in the study population rather than the policy. Stable support with negligible oracle headroom means that the frozen candidate generator, horizon, and metric expose little exploitable non-myopic structure; it does not imply universal myopia.

If meaningful headroom exists, actor-visible Q_1 tests whether the declared information state contains enough immediate target signal. Failure there directs attention to target association, leakage, representation support, or calibration before long-horizon architecture. Passing Q_1 but failing exact Q_2 localizes the first non-myopic defect to successor coverage, state aliasing, recursive targets, or bootstrap estimation. Only after those gates pass does failed endpoint recovery become evidence about planning under the admitted model and replay support.

Even then, the endpoint estimate alone does not identify which representation or optimization mechanism failed.

This logic also disciplines positive outcomes. Exact Q_2 would validate the first recursive prediction on its admitted support, not the complete policy. Endpoint recovery would remain conditional on the target protocol, finite candidate generator, hard-validity regime, horizon, oracle metric, and ASE population. The privileged bounded-lookahead policy is a reference within that finite experimental world, not a representation-independent or deployable upper bound.

7.2 Systems and external-validity boundaries

Counterfactual scoring repeatedly renders a large mesh, so memory and latency constrain branch factor and rollout volume before statistical efficiency can be assessed. An out-of-memory event motivates batching and resource measurement; it says nothing about candidate utility or planning value. Completed validated stores are required before throughput, storage, failure rates, or population coverage can be generalized.

The offline ASE setting additionally supplies geometry and target identity that a deployed egocentric actor would have to infer. The thesis therefore separates oracle task construction from actor-visible scoring end to end: target instruction, candidate proposal, hard mask, selected observation, and scorer input. Real-device or continuous-control claims would change observation, action, safety, and cost assumptions and remain outside the present evidence.

The conceptual contribution is consequently an identifiability structure for target-specific view planning. View value is relational, invalidity is distinct from utility, privileged support is distinct from actor visibility, and each scientific claim is attached to the earliest evidence gate capable of falsifying it. That structure remains useful whether the eventual policy result is positive, negative, or blocked.

8 Conclusion

This thesis formulates target-specific egocentric next-best-view planning as a relational finite-action problem: value depends jointly on target, causal information state, admissible candidate support, horizon, update rule, and continuation policy. It contributes a target-cropped reconstruction outcome, an explicit actor–oracle information boundary, a strict separation of feasibility from utility, causal selected-transition replay, a scalar-horizon fitted-value method, and a sequential evaluation from measurement validity to endpoint recovery.

The current evidence does not yet answer RQ2. Metric repeatability, the held-out target and action population, and actor-visible immediate-value recovery remain unestablished; without those premises, neither oracle-lookahead headroom nor learned exact-Q_2 and endpoint recovery are interpretable. Online interaction and continuous control remain extensions of the setting rather than missing parts of this evaluation.

Training-source rollouts establish that the pipeline reaches mesh rendering, target-specific oracle scoring, selected-action replay, and the fitted-value interface; renderer memory limits the evaluated scale. These observations justify an auditable method and study design, not policy superiority, a population effect, deployment readiness, or a scale estimate.

The eventual conclusion is determined by the first failed gate. Unstable measurement blocks planning claims; inadequate support blocks population claims; negligible headroom is a setup-specific negative result; failed actor-visible Q_1 or exact Q_2 localizes a learning prerequisite; and stable headroom without endpoint recovery is a bounded learned-policy failure. This ordering preserves informative negative outcomes without extending them beyond the frozen finite-candidate experiment.

Development roadmap

This page records schedule, status, evidence pointers, and promotion gates. It is omitted from submission output. The active thesis owns claims; Python, tests, configuration, and immutable artifacts own behavior and measurements.

Implementation: partial **Evidence: pending**

Planning status only; this view promotes no policy-improvement claim.

Promotion gate: M1 correctness and M5 headroom evidence

Development source: `docs/typst/thesis/sections/`; `docs/typst/thesis/development/m1-contract-report.typ`; `aria_nbv/`; `aria_nbv/tests/`; active configuration

Outcome

The development outcome is a reproducible, evidence-backed thesis package. Canonical narrative and equation labels live in `docs/typst/thesis/sections/`; this page tracks readiness and gates only.

Milestones

- **M0 — scope and foundation (2026-04-29–2026-05-10): status: complete** gate: thesis scope, source policy, and citation owners aligned. Evidence: `docs/typst/thesis/main.typ`, `docs/typst/thesis/sections/01-research-questions.typ`, `docs/references.bib`.
- **M1 — data, cache, and oracle correctness (2026-05-11–2026-05-31): status: blocked** gate: fresh store, scene split, frame/depth, candidate alignment, Rerun, and throughput evidence. Evidence: `docs/typst/thesis/development/m1-contract-report.typ#ssec:m1-status`, `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, `aria_nbv/tests/rollouts/`, `.configs/offline_only.toml`.
- **M2 — VIN baseline and scale gate (2026-06-01–2026-06-21): status: blocked by M1** gate: reproducible baseline, diagnostics, and scale receipt. Evidence: `aria_nbv/tests/vin/`, `aria_nbv/tests/lightning/`, `docs/typst/thesis/sections/05-experimental-design/05-01-objectives-and-hypotheses.typ`, `docs/typst/`

thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ.

- **M3 — target-RRI and generation readiness (2026-06-22–2026-07-12): status: blocked by M1** gate: trusted target subset, actor-visible protocol, and deterministic generation checks. Evidence: docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ, aria_nbv/tests/oracle/test_target_selection.py, aria_nbv/tests/pose_generation/, aria_nbv/tests/rollouts/.
- **M4 — target-conditioned one-step scorer (2026-07-13–2026-08-09): status: pending M3 evidence** gate: held-out ranking, calibration, oracle selections, and grouped failures. Evidence: docs/typst/thesis/sections/04-method/04-02-descriptor-and-encoding-plan.typ, aria_nbv/tests/oracle/test_scoring.py, aria_nbv/tests/lightning/test_candidate_scorer_contract.py.
- **M5 — lookahead headroom and Q_H (2026-08-10–2026-08-30): status: pending prerequisite gates** gate: equal-budget rollout comparisons, measured headroom, and supported Q_H evidence. Evidence: docs/typst/thesis/sections/04-method/04-05-finite-candidate-value-model.typ, docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ, aria_nbv/tests/data_handling/test_qh.py, aria_nbv/tests/rollouts/.
- **M6 — scale and bridge decisions (2026-08-31–2026-09-13): status: pending M5** gate: scale decision and explicit bridge scope. Evidence: docs/typst/thesis/sections/07-discussion.typ, aria_nbv/tests/oracle/test_online_vin.py, aria_nbv/tests/rollouts/test_reporting.py.
- **M7 — final experiments and writing (2026-09-14–2026-09-27): status: pending M5/M6** gate: immutable run/config links, coverage statement, final tables, and failure cases. Evidence: docs/typst/thesis/sections/06-results.typ, docs/typst/thesis/sections/07-discussion.typ, aria_nbv/tests/rollouts/test_reporting.py.

- **M8 — release freeze (2026-09-28–2026-09-30): status: pending M7** gate: reproducible configs, docs, demo path, smoke checks, and completed HM/FK07 pre-submission checks. Evidence: `Makefile`, `docs/README.md`, CI workflows, final evidence bundle, and authenticated PRIMUSS registration records.

Ablations

Comparison axes remain in canonical owners; this page records pointers and gates only.

- **Target input:** `docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ`; gate: V0/V1 protocol tests and actor-visible boundary.
- **Candidate mixture:** `docs/typst/thesis/sections/03-oracle-and-data-generation/03-02-target-task-and-rri-labels.typ`, `aria_nbv/aria_nbv/pose_generation/`; gate: validity and provenance diagnostics.
- **Objective/planner:** `docs/typst/thesis/sections/05-experimental-design/05-01-objectives-and-hypotheses.typ`, `docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ`; gate: equal budget and oracle receipt.
- **Invalidity/learning:** `docs/typst/thesis/sections/04-method/04-04-architecture-contract.typ`, `docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ`; gate: masks, reason codes, replay tests.
- **State/scale:** `docs/typst/thesis/sections/04-method/04-01-scene-representation-requirements.typ`, `docs/typst/thesis/sections/03-oracle-and-data-generation/03-03-replay-stores-and-diagnostics.typ`; gate: support and coverage accounting.

Evidence

- M1: `docs/typst/thesis/development/m1-contract-report.typ`, focused data/geometry/rollout tests, `.configs/offline_only.toml`, immutable store/Rerun receipts.

- M2–M4: `aria_nbv/tests/vin/`, `aria_nbv/tests/oracle/`, `aria_nbv/tests/lightning/`, and `docs/typst/thesis/sections/05-experimental-design/` plus saved calibration/ranking artifacts.
- M5–M7: rollout manifests, reports, configs, result tables, and exact support/coverage gaps in those artifacts.
- M8: CI, `make qmd-frontmatter-check`, `make thesis-pdf`, final smoke receipts, authenticated registration records, and the HM/FK07 pre-submission checklist; tracked PDF remains a development artifact and cannot establish submission.

Risks

Active risk state is pointer-only: frame/depth `aria_nbv/tests/rendering/test_depth_backprojection_conventions.py`, target validity `aria_nbv/tests/oracle/test_target_selection.py`, rollout support `aria_nbv/tests/rollouts/test_zarr_store.py`. Open blockers remain in `docs/typst/thesis/development/m1-contract-report.typ` and `.agents/issues.toml`/`.agents/todos.toml`; no duplicate risk narrative is maintained here.

Freeze

Freeze gate: clean reproducible smoke matrix, immutable run/config links, final coverage statement, and no stale paths or placeholders. The exact submission evidence gate remains owned by `docs/typst/thesis/main.typ` and its report bundle; this development projection does not satisfy it.

HM/FK07 release checks are human-owned and must be completed against the author's authenticated records and the final PDF at the applicable pre- or post-submission stage:

- **Applicable rules:** confirm the individually assigned programme and SPO version in PRIMUSS; public reading versions establish only the current baseline.
- **Registration:** retain the PRIMUSS confirmation, official start date, individual deadline, assigned examiners, supervisor-specific length and form, and the attachments requested for this thesis.

- **Pre-submission readiness:** verify the final upload, mandatory attachments, and exact PDF prepared for PRIMUSS upload before the individual deadline. A repository field, local build, PDF digest, or upload alone is not submission evidence.
- **Copies:** provide a paper copy only when an examiner requests one under the current FK07 process.
- **Local final-PDF review:** as a repository/author release check, verify the declaration and AI/tool disclosure in the exact PDF prepared for PRIMUSS upload. Confirm any PRIMUSS-marked mandatory declaration, signature, or attachment against the authenticated record and assigned supervisor; this checklist does not impose a universal signature form.
- **Post-freeze submission closure:** at the actual submission, invoke `Abschlussarbeit abgeben`, retain the authenticated timestamp, and archive the exact submitted PDF. This later receipt closes the submission evidence gate but does not block the September M8 development freeze.

Official baseline checked 2026-08-25: HM ASPO, Master Informatik SPO, FK07 thesis process, PRIMUSS registration guide, and PRIMUSS submission guide. Recheck the official versions and authenticated instructions at freeze time.

Promotion queue

Promotion queue — candidate: Promote M5 lookahead/Q_H results after paired held-out evidence is immutable.
Source: docs/typst/thesis/sections/05-experimental-design/; aria_nbv/tests/rollouts/test_reporting.py target: docs/typst/thesis/sections/06-results.typ and docs/typst/thesis/sections/07-discussion.typ gate: positive headroom, oracle re-scoring, uncertainty, and coverage

Promotion queue — blocked: Resolve the scale fallback before claiming population coverage.
Source: docs/typst/thesis/development/m1-contract-report.typ; aria_nbv/tests/data_handling/; aria_nbv/tests/rollouts/ target: docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ gate: throughput and scene-level held-out coverage

Promotion queue — deferred: Keep online discrete Q_H and continuous target-then-pose work as bridge scope until offline gates close.
Source: docs/typst/thesis/sections/07-discussion.typ; .agents/issues.toml target: docs/typst/thesis/sections/07-discussion.typ gate: stable offline Q_H and explicit scope decision

Promotion queue — rejected: Do not promote unsupported continuous, VLM, semantic-global, or real-device claims into the thesis core.

Source: docs/typst/thesis/main.typ; docs/typst/thesis/sections/07-discussion.typ target: docs/typst/thesis/sections/08-conclusion.typ gate: bounded non-claim retained

Priorities

Gate order is M1 correctness, M2–M4 target/scorer readiness, M5 headroom, M6 bridge decision, then M7/M8 freeze. Each transition requires the evidence pointer and exit gate above; no scale or bridge work bypasses a blocked prerequisite.

Issues and blockers

- M1 store, scene split, Rerun, and throughput: `docs/typst/thesis/development/m1-contract-report.typ#m1-blockers`.
- M5 support and headroom: `docs/typst/thesis/sections/05-experimental-design/05-02-learning-objective-and-replay-evidence.typ`, `docs/typst/thesis/sections/05-experimental-design/05-03-policy-comparison-and-failure-interpretation.typ`.
- Actionable follow-up: `.agents/issues.toml` and `.agents/todos.toml`; these records do not replace canonical owners.

M1 Contract Report

Status

Implementation: partial **Evidence: pending**

Synthetic contract guards are available, but the configured local store and scale evidence do not yet close the M1 exit gate.

Promotion gate: fresh store, scene-split, Rerun, and throughput evidence

Development source: thesis M1 gate notes and executable package owners (not a second contract owner)

Area	Status	Owner and evidence pointer
Source-contract guards	validated	Focused tests under `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, and `aria_nbv/tests/rollouts/` own the executable checks; this view records only their M1 status.
Configured local store	blocked	`.configs/offline_only.toml` and the resolved `.data/offline_cache/vin_offline` artifact are the smoke-evidence owners.
Scene-level split evidence	blocked	The store artifact and its split metadata require a scene-level train/validation proof before promotion.
Rerun diagnostics	blocked	No M1 Rerun recordings are present in this worktree. The recording gate remains open; absence is reported explicitly rather than linked to a nonexistent artifact.
Oracle throughput	blocked	Representative mesh, rendering, backprojection, and RRI timing remains to be captured in the experiment evidence owner.

Evidence

The current evidence view is intentionally pointer-only. The defining Python modules, tests, active configuration, and generated artifacts remain authoritative; no schema, tensor shape, formula, serialized value, or command ledger is reproduced here.

- Synthetic guard results: `aria_nbv/tests/data_handling/`, `aria_nbv/tests/pose_generation/`, and `aria_nbv/tests/rollouts/`.
- Local smoke configuration and artifact root: `.configs/offline_only.toml` and `.data/offline_cache/vin_offline/`.

- Rerun evidence: absent in the current worktree; the normal/boundary/failure recording gate remains open.

Blockers

Promotion queue — blocked: Refresh the configured-store evidence and report coverage status.

Source: M1 gate notes and executable package owners target: M1 status and evidence gate: fresh immutable-store inspection

Promotion queue — blocked: Replace the one-scene diagnostic split with scene-level evidence or record the exact residual blocker.

Source: `.data/offline_cache/vin_offline/` target: M1 evidence gate: scene-level split proof

Promotion queue — blocked: Collect normal, boundary, and failure Rerun recordings.

Source: `.artifacts/rerun/` target: M1 evidence gate: three diagnostic recordings

Promotion queue — blocked: Measure representative oracle throughput and preserve the measurement artifact for review.

Source: `aria_nbv/` and active experiment configuration target: M1 evidence gate: representative timing receipt

9 Development Pilots

9.1 Candidate-Family Support

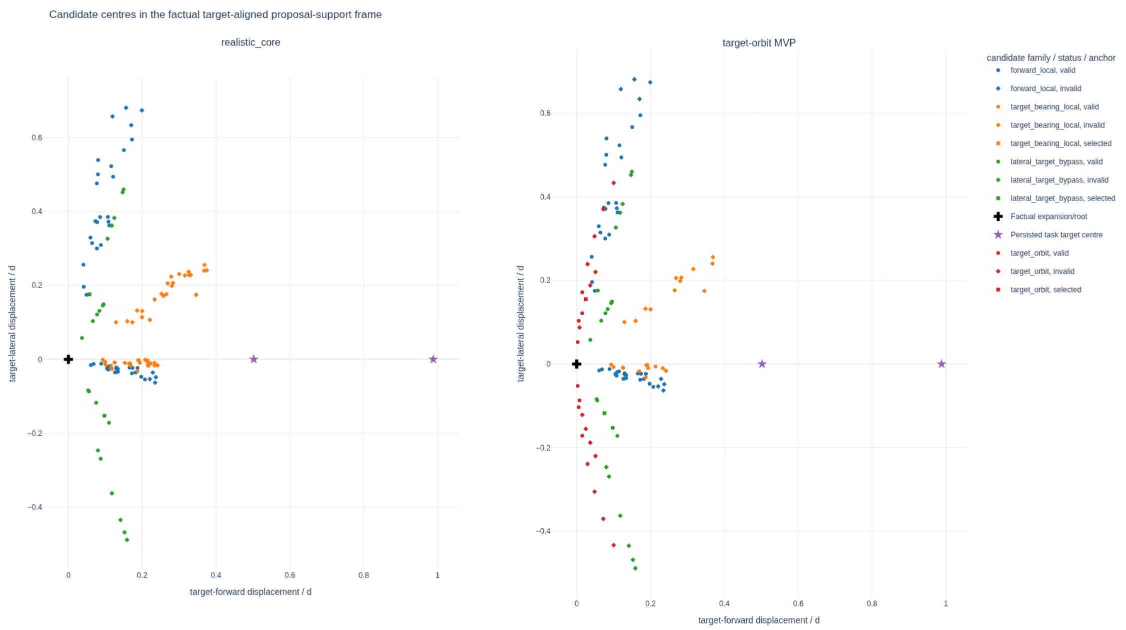

Figure 9: Attempted candidate centres from a two-scene, two-state real-data method audit. Each state is centred on its factual expansion pose, yaw-aligned to the oracle-evaluation target with world Z-up, and divided by its current three-dimensional target distance. The cross marks the expansion/root pose, the stars mark the two oracle task target centres, colour denotes proposal family, and marker shape denotes actor-valid or selected status. The figure diagnoses proposal geometry under the `v0_gt_input` target protocol; it is neither a population estimate nor evidence of actor-visible target discovery or policy quality.

The target-orbit challenger adds attempted support on both target-relative sides in this bounded pilot, while pruning still leaves invalid rows and one configured family/state pair without actor-valid support. The associated immutable evidence manifest records the common-frame reducer, store-manifest hashes, expected state index, committed reduced candidate rows, resolved challenger profile, reproduction commands, metric denominators, and plot hashes. No frame-mixed side-balance or orbit-span values are used. The interactive evidence and Streamlit view can optionally add short arrows for the valid candidates' camera optical axes; the static thesis figure omits them to preserve legibility.

9.2 Target-Frame $\mathcal{S}^2$ Diagnostics

The following frozen figures are generated by the same Python reporting and plotting owners as the stored-rollout application. For the selected real-data shard, the report records these support counts: source samples 1, snippets 1, scenes 1, targets 1, retained rollout chains 1, and selected steps 5. It remains pilot evidence: its purpose is to validate the representation and reporting contract, not to estimate policy quality. The two point-direction views contain 5 factual displacements and 5 factual optical axes. Colour denotes rollout-chain identity, marker shape denotes decision-step identity, and the sphere heat map is computed from the complete equal-solid-angle count grid rather than from the bounded incidence overlay.

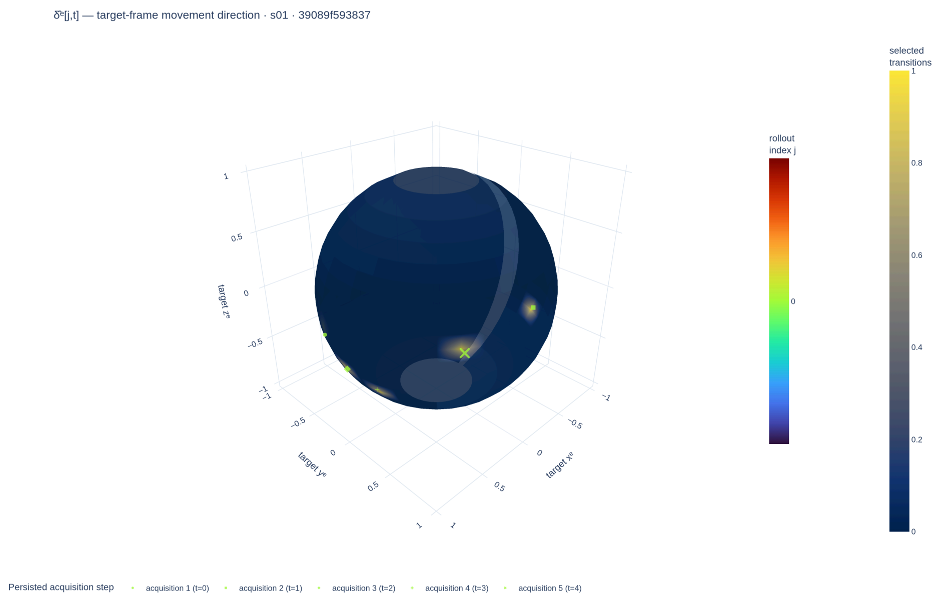

Figure 10: Target-frame $\mathcal{S}^2$ movement directions from the frozen pilot report. This view records how the camera moved; it is not a visibility measurement.

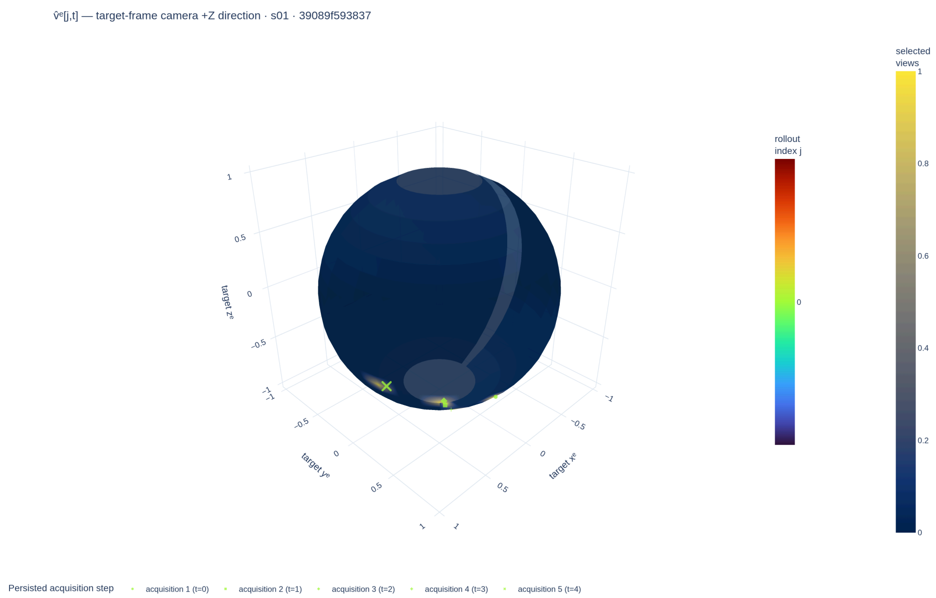

Figure 11: Target-frame $\mathcal{S}^2$ selected-camera optical-axis directions from the same frozen pilot report. This view records where the camera pointed; it is distinct from both camera motion and calibrated surface support.

The calibrated proxy diagnostic integrates 5 selected frusta. Their mean intrinsic field-of-view solid angle is 2.49 sr. On the geometric-mean-scale proxy sphere, one view covers a mean fraction of 0.168, while their union covers 0.312. These fractions express geometric potential support under the proxy and front-facing tests; they exclude true target shape, depth ordering, and scene occlusion.

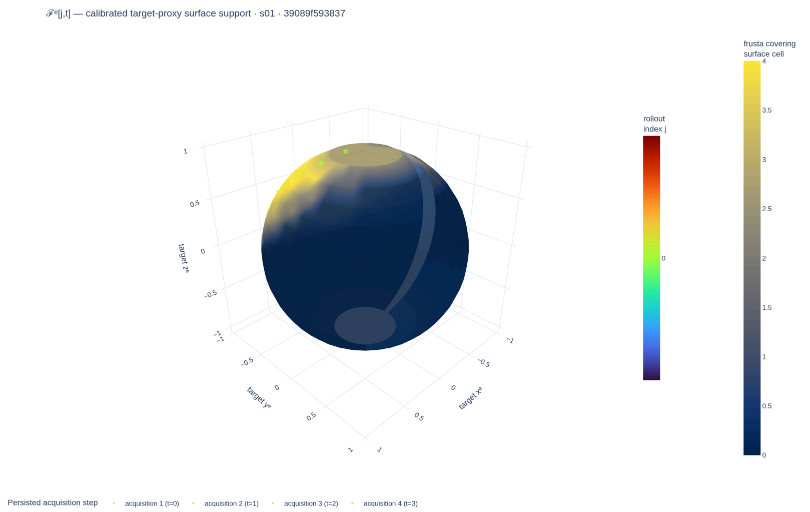

Figure 12: Calibrated selected-frustum support on the target-centred proxy sphere. The surface heat map is the complete equal-solid-angle coverage field; overlaid incidence samples preserve rollout and time heritage. This is not measured target-mesh visibility.

Development Register: Method Alternatives

This register preserves executable but unpromoted controls and future hypotheses outside the thesis-core Method chapter. Entry here means that an idea has a named comparison and promotion gate; it does not mean that the thesis evaluates or recommends it.

Implementation: exploratory **Evidence: pending**

The thesis-core selection remains A1–H0–S0 with direct continuous regression. Every item below is excluded from its central method claim.

Promotion gate: promote only a one-factor comparison with an immutable receipt and untouched scene-disjoint evaluation

Development source: `aria_nbv/aria_nbv/vin/`; `aria_nbv/aria_nbv/lightning/`; `docs/contents/theory/`

State and history alternatives

- **S1 selected surfaces:** the executable privileged control backprojects only factual selected mesh depth, pools the causal point set, and initializes its residual at zero so a fresh S1 scorer matches H0. It lacks free/unknown evidence, is density weighted, and has no confirmatory receipt; it therefore cannot support a geometry-benefit or deployment claim.
- **S2 ray-aware memory:** a planned state would distinguish observed surface, observed free, unknown, support, uncertainty, source, and recency. It requires deterministic fusion and no-future-observation tests before a model comparison.
- **H1 ordered history:** an executable causal Transformer adds relative age to the same selected-pose tokens used by H0. It remains a pose-only capacity ablation until repeated-seed endpoint evidence justifies chronology.
- **Other carriers:** target-centred EFM3D re-lifting, appearance-on-point, sparse TSDF/SDF encoders, object-aware 3D Gaussian splats, and SceneScript context are hypotheses for diagnosed representation failures, not parallel thesis methods.

Interaction alternatives

A2 DeepSets context, A3 masked candidate self-attention, A4 query-local relation bias, A5 recurrent state reading, A6 residual value prediction, and A7+ graph or exact-equivariant layers are ordered escalation points. A level may enter the main method only after A0/A1 are measured under the same state, target protocol, decoder, objective, and evaluation contract and a concrete failure identifies the missing interaction. Candidate-set models must retain row equivariance, invalid-row isolation, duplicate tests, and absolute-value semantics.

Objective and estimator alternatives

- **CORAL decoding** is executable over the same continuous fitted-Q targets, but its ordinal loss and fixed representatives add support-provenance and saturation questions. It is excluded until direct regression supplies the frozen control.
- **Fixed-horizon or separate heads** alter parameter sharing and the public interface. They are ablations against the scalar requested-horizon family, not co-equal definitions of Q_H.
- **Behavior-return Monte Carlo regression** estimates the continuation policy that generated each chain rather than greedy finite-support value unless those policies coincide. It must retain that distinct estimand.
- **One-step plus residual prediction** can test whether calibrated immediate utility is a useful base, but the residual may not be mean-centred across a candidate table because unrelated or duplicate rows would then redefine an action's absolute target.
- **Quantile regression** is justified only after stochastic return variation or state aliasing is measured and a distributional Bellman projection, support, decoding rule, and calibration protocol are frozen.

The promotion sequence is: establish population/action support, measure oracle headroom, learn actor-visible Q_1, pass exact Q_2, and only then introduce the smallest alternative that addresses the observed failure.

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A Reproducibility Record

The report bundle is the single numerical interface between validated rollout artifacts and this manuscript. Its schema version, evidence status, store manifests, typed parameter rows, failure records, and sidecar hashes preserve provenance without duplicating analysis logic in Typst. The document loader rejects schema or status mismatches before any table is rendered. Compact labels and digest prefixes are used below; complete store identifiers, paths, and hashes remain in the machine-readable bundle.

The implementation anchors behind that interface are the immutable `vin_offline/` source store, the task-specific `rollouts.zarr/` replay store, and the derived `q_h/` training cache. Their exact schemas, normalized joins, chunking, codecs, receipt versions, and execution commands remain manifest-owned reproducibility metadata rather than scientific entities in Chapter 3. A reported row is admissible only when the bundle resolves these identities and preserves source role, missingness, and failure records.

Evidence class	Store	Manifest digest	Validated
Development fixture	store S1	not-a-real-m…	yes

Table 12: Loaded development-fixture provenance. The fixture verifies the document interface only and is not scientific evidence.

The bundled development fixture contains synthetic values solely to test typed loading and missingness. Its numerical rows are deliberately not rendered or interpreted.

B Development Diary

Archived source note: This marked diary is a development-only record. It is excluded from submission mode; only evidence-backed definitions, protocols, and results graduate into the thesis body.

Source: thesis mode contract

B.1 Decisions Retained

The thesis keeps the privileged oracle and actor-visible policy as separate contracts, evaluates finite candidate sets under a shared validity mask, and treats target-specific reconstruction improvement as the common comparison signal. Rollout stores retain selected transitions and provenance rather than becoming a second implementation of the oracle or training pipeline.

Open decision: Freeze the scene-disjoint analysis population and the minimum evidence scale only when the resolved manifests and exclusion ledger are available.

Source: experimental-design contract

Gate: confirmatory bundle freeze

B.2 Failures That Changed the Plan

Early rollout attempts reached the mesh-rendering path but exceeded available accelerator memory when candidate views were rendered without a bounded batch. This failure narrowed the immediate claim to implementation readiness and made peak memory, throughput, failure provenance, and completed-store validation explicit scale gates.

Validation TODO: Require completed stores, matching manifests, resource summaries, and recorded failures before extrapolating rollout bandwidth or storage demand.

Source: rollout attempt artifacts

Gate: cluster-readiness evidence

B.3 Rejected Scope

Continuous actions, online simulators, additional scene representations, and alternative reinforcement-learning families are excluded from the core study while the finite-candidate target-conditioned comparison remains unvalidated. They would change both the action contract and the supervision regime, so adding them now would weaken rather than extend the primary experiment.

Remove or rewrite before submission: Delete bridge material that cannot be tied to a concrete limitation or a matched follow-up experiment after the confirmatory analysis is complete.

Source: scope review

Gate: discussion freeze

B.4 Directional-Memory Hypothesis

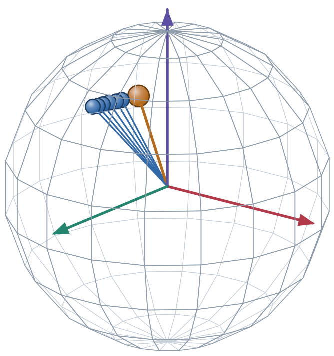
Research TODO: Treat target-centred directional memory as an ablation hypothesis, not an actor feature or contribution. The representation must be implemented, mask-audited, and compared against pose distance and overlap before any predictive claim enters the manuscript.

Source: RQ3 representation hypothesis

Gate: paired held-out ablation

A. Target-centred directions on $\mathbb{S}^2$

actual logged camera centres, expressed in the target-object frame

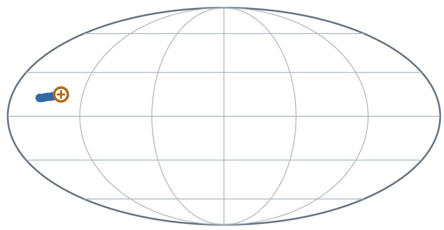


● logged history direction ⊕ example query

Scene 81286, sample ASE_81286_Atek_000024, target instance 128 frames 0, 2, 4, 6, 8, 10 and query frame 15.

B. Complete-domain view and prospective compression

Mollweide is equal-area; no smooth field or SH lobe is implied



$+x_e$ at map centre · $+z_e$ north · wrap seam at $\pm 180°$

$$\boldsymbol{M}_{\mathrm{dir}(v)} = \textstyle\sum_{k<t} w_{k(v)} \boldsymbol{d}_{k(v)} \boldsymbol{d}_{k(v)}^{\top}$$

$$\nu_{i(v)} = 1 - \frac{\boldsymbol{d}_i^{\top} \boldsymbol{M}_{\mathrm{dir}} \boldsymbol{d}_i}{\mathrm{tr}\, \boldsymbol{M}_{\mathrm{dir}} + \varepsilon}$$

This fixture uses uniform weights and a logged future pose only to make the second-moment query inspectable. It does not claim that the learner currently stores $\boldsymbol{M}_{\mathrm{dir}}$ or that this novelty predicts target RRI.

HYPOTHESIS / ABLATION MATERIAL. Keep outside submission claims until the representation exists and paired evaluation tests whether it adds information beyond pose distance and overlap.

Figure 13: Development-only directional-memory fixture. The sphere and Mollweide panels show the same logged ASE camera directions in a target-object frame; the orange query is another logged pose used to inspect one prospective second-moment novelty definition. No smooth field, spherical-harmonic representation, or learned benefit is claimed.

B.5 Next Evidence Gates

The next admissible evidence is a validated report bundle that fixes the study population, records target and candidate exclusions, supplies paired endpoint outcomes with uncertainty, and joins runtime and storage statistics to the same manifests. Only then can the manuscript replace the present availability statements with policy comparisons.

Implementation TODO: Export the confirmatory report bundle, compile both thesis modes against it, and inspect the resulting tables and failure cases before freezing claims.

Source: reporting contract

Gate: claim freeze

Ehrenwoertliche Erklaerung

Ich versichere, dass ich diese Masterarbeit selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen oder Hilfsmittel benutzt sowie wörtliche oder sinngemäße Zitate als solche gekennzeichnet habe.

Muenchen, 7. Jan 2027 Jan Duchscherer